
SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

Anonymous authors
Anonymous submission

ABSTRACT

Spatial reasoning is a fundamental aspect of human cognition, yet current vision–language models (VLMs) still show large, measurable deficits when spatial judgments are evaluated in realistic scenes. Prior benchmarks have often emphasized synthetic or LLM-generated environments, narrow task formats, and puzzle-like setups, making it difficult to quantify performance under the visual noise, object clutter, and heterogeneous spatial relations that VLMs face in deployment. To support clearer empirical comparison, we introduce **SPATIALAB**, a real-world benchmark for measuring VLM spatial reasoning under unconstrained visual conditions. SPATIALAB contains 1,400 visual question–answer pairs distributed across six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*. Each category is decomposed into five subcategories, producing 30 task types with at least 25 questions per subcategory and at least 200 questions per main category. This balanced design supports direct comparison across methods, categories, and evaluation formats through both multiple-choice and open-ended protocols. Across a diverse model suite spanning open-source, closed-source, reasoning-focused, and spatial-specialist VLMs, the benchmark reveals consistent empirical gaps relative to human performance. In multiple-choice evaluation, the strongest model, InternVL3.5-72B, reaches 54.93% accuracy compared with 87.57% for humans. In open-ended evaluation, all models drop by roughly 10–25% relative to their multiple-choice scores; GPT-5-mini leads this setting with 40.93% accuracy, still well below the 64.93% human baseline. The largest and most persistent deficits appear in complex spatial relations, depth perception, navigation, and 3D geometry, indicating that current gains remain task- and format-dependent. By pairing a balanced real-world taxonomy with human baselines and per-format evaluation, SPATIALAB makes quantitative comparison of spatial reasoning more diagnostic. The resulting patterns identify where models are improving, where margins remain large, and which evaluation conditions expose practical failures in robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is a core component of human cognition: it enables people to parse layouts, mentally transform objects, and navigate physical environments. It also underpins high-impact applications in robotics, autonomous driving, augmented and virtual reality, geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents operating in dynamic scenes, useful spatial competence must be measured not only by semantic recognition but also by performance on geometric relations, depth cues, occlusion patterns, and agent-centric perspectives in visually noisy and structurally complex environments (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Although recent vision–language models have made strong gains in multimodal representation and language grounding, their measured spatial accuracy remains fragile when evaluation shifts to realistic scenes (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Existing spatial benchmarks provide useful controlled comparisons, but their quantitative signal is often concentrated in narrow and simplified conditions. Many focus on binary spatial relations, coarse depth categories, or schematic navigation cues, frequently using synthetic or simplified

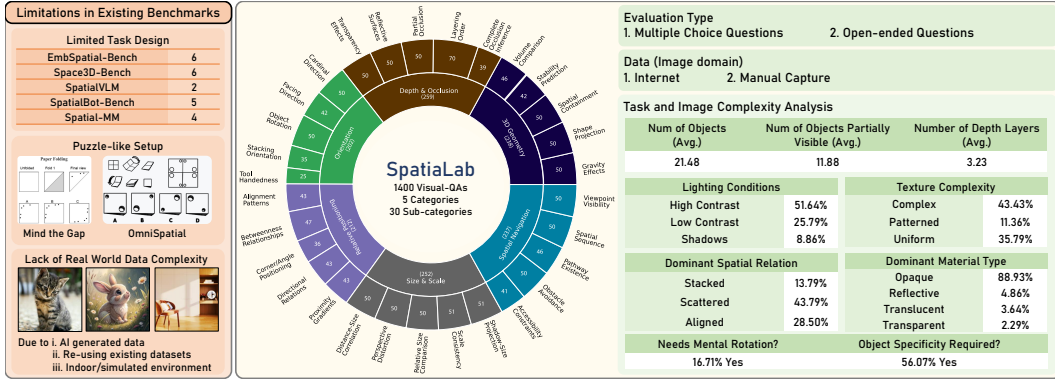


Figure 1: **Overview of SPATIALLAB.** The benchmark turns the comparison structure explicit: it contrasts prior dataset limitations (left), evaluates 1,400 visual QA pairs across six categories and 30 subcategories (center), and supports both multiple-choice and open-ended scoring. The image set is designed to make empirical differences visible across object counts, depth layers, lighting, textures, relations, and materials (right).

scenes that reduce clutter and homogenize object appearance (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). Such settings can make model scores appear saturated while leaving distribution-shift failures under-measured. The categories most relevant to deployment—occlusion inference, scale-consistent judgments across viewpoints, and planning under partial observability—remain comparatively sparse (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). Consequently, high scores on existing benchmarks do not necessarily predict reliable multi-step spatial inference, motivating an evaluation where performance margins are measured in real-world scenes and reported across fine-grained spatial skills.

A meaningful spatial benchmark must therefore measure all core axes of spatial reasoning under the same evaluation umbrella: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. Human cognition integrates these axes under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025), so model evaluation should report not only aggregate accuracy but also how accuracy changes by category, answer format, and comparison baseline. Multiple-choice questions quantify discriminative selection under controlled distractors, open-ended questions quantify generative grounding, and human baselines make the absolute performance margin interpretable. These principles guide our benchmark design.

To meet these needs, we introduce **SPATIALLAB**, a benchmark for comparing VLM spatial reasoning in realistic, unconstrained visual contexts, as outlined in Figure 1. **SPATIALLAB** provides a balanced suite of visual question-answer items, a broad taxonomy of spatial tasks, and paired multiple-choice and open-ended formats so that method, metric, dataset, and evaluation condition are explicit in every comparison. The benchmark emphasizes photographic diversity, complex relational queries, and tasks that require integrating depth, perspective, and 3D structure. We evaluate a broad set of state-of-the-art VLMs, including open-source, closed-source, reasoning-focused, and spatial-specialist systems, compare each group against human performance, and analyze the error patterns behind the observed margins. Our key contributions are as follows:

1. We introduce **SPATIALLAB**, a 1,400-item benchmark spanning six principal categories and thirty task types in realistic, unconstrained contexts. With at least 200 questions per category and 25 per subcategory, it provides balanced coverage for both multiple-choice (SPATIALLAB-MCQ) and open-ended (SPATIALLAB-OPEN) evaluation.
2. We conduct a **comprehensive evaluation** of 25+ state-of-the-art vision-language models—including open-source, closed-source, reasoning-oriented, and spatial-specialist systems—against human baselines, making the resulting accuracy margins and format-dependent gaps directly comparable.

Table 1: Comparison of spatial reasoning benchmarks by domain, task design, evaluation format, and reported performance.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval			A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.		Type	Depth					
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I		-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I		LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I		XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I		Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I		RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I		GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I		GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I		GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I		LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I		GPT-4V	68.00
VSI-Bench	Y	8	Indoor	M,E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V		Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I		o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I		InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix		Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix		InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix		GPT-5-mini	40.93

3. We provide **in-depth error analyses and diagnostic studies** that quantify where failures concentrate, especially in depth perception, navigation, occlusion, and 3D geometry, and translate these empirical patterns into actionable directions for robust spatial understanding.
4. We compare multiple enhancement strategies, including **supervised fine-tuning (SFT)**, **chain-of-thought (CoT) prompting**, **CoT with self-reflection**, and **multi-agent architectures (SPATIOXOLVER) with multi-step reasoning**, reporting where they improve accuracy and where gains fail to transfer across multiple-choice and open-ended settings.

The empirical picture is consistent across formats. In the multiple-choice setting, leading models cluster around 50–55% accuracy, whereas human annotators exceed 85%. In open-ended generation, models lose an additional 10–25 percentage points, with the strongest systems near 40% compared with about 65% for humans. Error rates concentrate in occlusion inference, scale consistency, and multi-step navigation, showing that the practical gap is not a single aggregate number but a repeated pattern across categories and answer formats. By releasing tasks, baselines, and diagnostics together, SPATIALAB establishes a reproducible comparison point for measuring spatial reasoning progress.

2 PRELIMINARIES: SPATIAL REASONING IN VLMS

Visual-spatial reasoning is the ability to perceive, represent, and manipulate spatial relations among objects, and it is central both to human cognition and to embodied agents that must navigate and act in the world (Palmer, 1999; Chabris et al., 2006). Formally, we can describe a scene as a set of objects $O = \{o_1, \dots, o_n\}$ with attributes (e.g., position, orientation, size) in a world coordinate system W . Visual-spatial reasoning then seeks to learn a mapping $f : (I, Q) \mapsto R$, where I is an image (or image sequence), Q is a linguistic query, and R is a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ capturing predicates such as *left-of*, *occludes*, or *supports*. For VLMS, the empirical difficulty of learning f comes from jointly optimizing visual perception and compositional linguistic abstraction under perspective, gravity, and physical-interaction constraints (Dai et al., 2017; Alayrac et al., 2022).

Prior VLM spatial-reasoning evaluations have typically measured a limited subset of this mapping. Many rely on basic spatial relations or narrow task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), often in synthetic datasets such as CLEVR (Johnson et al., 2016), structured QA benchmarks such as GQA (Hudson & Manning, 2019), or puzzle-like simulated setups such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025). Table 1¹ summarizes these comparison dimensions. The pattern is clear: controlled settings are valuable for isolating variables, but they often reduce environmental noise, underrepresent viewpoint shifts, and limit the quantitative evidence available for real-world generalization. We therefore benchmark spatial reasoning in richer contexts that expose models to the scene complexity and perspective variation expected in robotics, AR/VR, and autonomous navigation (Mu et al., 2023; Collaboration et al., 2023).

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

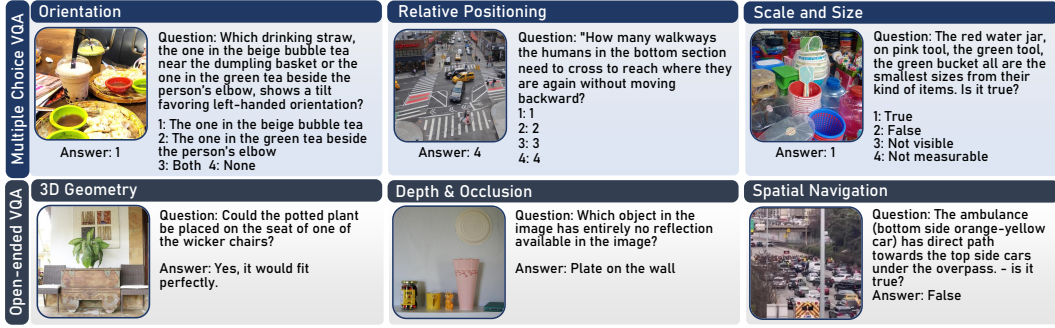


Figure 2: Representative SPATIALLAB examples from the six spatial categories, shown in both open-ended and multiple-choice formats to illustrate the paired evaluation conditions used for quantitative comparison.

In SPATIALLAB, spatial reasoning is organized into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each connected to cognitive faculties studied in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is decomposed into five subcategories, yielding thirty task types that isolate measurable skills such as directional alignment, occlusion inference, and stability prediction. This taxonomy makes category-level margins interpretable: a model can be compared not only by whether it solves the benchmark overall, but by which spatial dimensions consistently improve or remain fragile (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

We also use two complementary evaluation formats because format changes the evidence provided by a score. Multiple-choice evaluation measures recognition and discrimination among distractors, whereas open-ended evaluation measures free-form generation, compositional description, and calibration under less constrained output conditions (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Reporting both metrics lets us identify whether a method improves robust spatial reasoning or primarily benefits from answer options. By combining a principled taxonomy, real-world imagery, dual formats, and human baselines, SPATIALLAB supports fine-grained model comparison and a clearer quantitative account of progress toward robust spatial cognition.

3 SPATIALLAB BENCHMARK

SPATIALLAB is designed to make spatial-reasoning comparisons measurable across task type, answer format, and model family. It contains 1,400 validated visual question-answer pairs collected from realistic scenes where occlusion, perspective distortion, clutter, and noisy backgrounds co-occur. The benchmark is balanced across six major categories—*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*—with five subcategories per category, for 30 task types in total. Each subcategory contains at least 25 QA pairs and each top-level category contributes at least 200, which keeps per-category accuracy comparisons from being dominated by a small or uneven slice of the dataset. Because every item supports both multiple-choice and open-ended evaluation, SPATIALLAB measures discriminative accuracy and generative reasoning under matched visual conditions. Figure 2 shows representative examples from the six categories and highlights the breadth of evidence used in the empirical comparisons.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was curated with three complementary sources: (i) automated large-scale web crawling, (ii) targeted online retrieval for scene-specific cases, and (iii) manual snapshots from diverse indoor and outdoor environments (Figure 3). This multi-source pipeline increases coverage across object categories, environmental conditions, and spatial configurations. To quantify visual diversity, each image was profiled along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints*

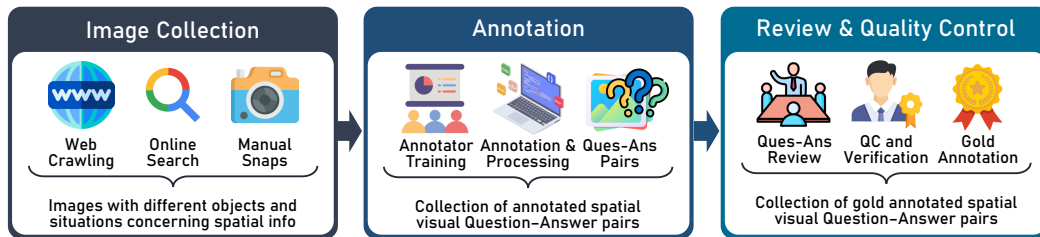


Figure 3: **Data creation pipeline for SPATIALAB.** Images are collected from web crawling, targeted search, and manual snapshots, then converted into structured spatial QA pairs and filtered through review so that the final 1,400 items support consistent MCQ and open-ended comparisons.

(normal, floating, and unconstrained). The final corpus systematically represents these dimensions, making the evaluation condition closer to in-the-wild deployment than to simplified or synthetic setups. Corpus statistics further characterize the empirical difficulty of the benchmark: images contain 21.48 objects on average, including 11.88 partially visible objects, across 3.23 distinct depth layers. Spatial reference chains (e.g., “object A is left of B, which is behind C”) average 2.07 links, indicating multi-step relational structure rather than isolated pairwise judgments. In addition, 16.71% of questions require mental rotation and 56.07% require object specificity such as color, pattern, or other fine details. These measurements show that **SPATIALAB** evaluates models on cluttered, layered, and relation-rich scenes that are underrepresented in many synthetic datasets.

Annotation. Annotation proceeded in three phases to keep the comparison set consistent across categories and answer formats. First, annotators completed targeted training (details are available in Appendix C.2) so that category and subcategory decisions followed shared criteria. Second, each image was paired with one or more spatial QA tasks covering reasoning types such as proximity gradients, alignment patterns, occlusion inference, and stability prediction. The questions balance perceptual grounding with higher-order inference, including cases such as predicting stability under gravity constraints. Third, each QA pair was encoded in both multiple-choice (04 options) and open-ended form, enabling matched evaluation of discriminative and generative performance. This process produced around 1,500 candidate spatial visual QA pairs across six categories and 30 subcategories.

Review and Quality Control. A three-tier review pipeline converted the candidate pool into a reliable evaluation set. The first stage checked semantic validity and subcategory alignment. The second stage used independent annotators to verify question phrasing and answer correctness, with emphasis on reducing ambiguity and enforcing task clarity. The third stage established a gold-standard annotation round for high-confidence evaluation items. All image metadata was removed, and the dataset was curated with attention to copyright concerns, although minor oversights may still occur. After review, the final benchmark contains 1,400 validated QA items, which are the basis for all reported quantitative comparisons.

4 EXPERIMENTS AND EVALUATION

Models. We evaluate models in four comparison groups so that performance can be interpreted by access type, scale, and reasoning specialization. **Proprietary models** include GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** comprise Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** comprise SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE, we report only open-ended evaluation because these models are not instruction-tuned and were trained primarily on open-ended tasks, which produced excessive refusals under MCQ prompting. For readability in the results tables, model groups are color

Table 2: Multiple-choice accuracy (%) (↑) on SPATIALAB-MCQ by spatial category and overall score. The table compares model families against the human baseline under the same MCQ evaluation condition.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

coded as follows: proprietary , open-source up to 7B , larger than 7B , proprietary reasoning , open-source reasoning , spatial specialists , and human baseline .

Evaluation Process and Metrics. Accuracy is the primary metric for both SPATIALAB-MCQ and SPATIALAB-OPEN, enabling direct comparison across model groups, spatial categories, and answer formats. For **MCQ evaluation**, models receive direct prompts and return the option number corresponding to their selected answer; responses are automatically matched against the gold answer. Detailed prompts and setup information are provided in Appendix D.1. For **open-ended evaluation**, models generate free-form answers that are scored by a large language model judge, with prompts and protocol details in Appendix D.2. To validate this scoring condition, we also measured LLM–human agreement, as detailed in Appendix D.2. Table 5 shows that the LLM judge (Gemini-2.5-Flash) achieves substantial agreement with human annotators: Cohen’s kappa is 0.738 against the majority vote and ranges from 0.681 to 0.795 against individual annotators. Raw accuracy against the majority vote is 0.880, while Fleiss’ kappa among human annotators is 0.774, supporting the reliability of the open-ended metric. For additional details on experiment setup, prompts, and evaluation procedures, see Appendix D.

Improving Visual Reasoning Capabilities. Beyond base-model comparison, we evaluate interventions that could shift spatial-reasoning accuracy under the same benchmark conditions: inherent reasoning in VLMs (Appendix H.1), Chain-of-Thoughts (CoT) (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). For SFT, Qwen2.5-VL-3B-Instruct is fine-tuned on a stratified 40% split of the dataset and evaluated on the remaining 60%, allowing gains to be measured on held-out relational and geometric tasks. The resulting pattern is informative: SFT reduces errors in navigation and orientation and can combine additively with CoT, whereas agentic reasoning yields strong orientation gains but stagnates or degrades in other categories. These comparisons show that improved accuracy is condition-specific and that deeper perceptual grounding remains necessary beyond multi-step reasoning scaffolds.

Table 3: Open-ended accuracy (%) (↑) on SPATIALAB-OPEN by spatial category and overall score. The table reports free-form answer performance judged under the same protocol, enabling direct comparison with MCQ results.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline	73.53	50.19	70.30	69.81	65.48	62.87	64.93

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 reports multiple-choice accuracy on SPATIALAB-MCQ, where the comparison is between each model group and the human baseline under the same 1,400-item evaluation. Model accuracy spans roughly 30–55% (random choice = 25%), while the human baseline reaches 87.57%, leaving a large absolute margin even for the strongest systems. The top model is InternVL3.5-72B, an open-source model, with 54.93% accuracy. The upper tier includes reasoning- and instruction-tuned models (o4-mini: 53.21%, Gemini-2.5-Flash-Thinking: 52.93%, GPT-5-mini: 54.29%) and large open-source models (InternVL3.5-72B: 54.93%, Llama-3.2-90B: 50.36%), but the ranking also shows that scale is not sufficient by itself: Llama-3.2-11B reaches only 30.50%. By category, Orientation and 3D Geometry provide the clearest high-performing pockets, with several strong models exceeding 60% on one of these dimensions. In contrast, Spatial Navigation, Depth & Occlusion, and Size & Scale remain more variable and generally harder across model families. Spatial-specialist models (SpaceOm, SpaceThinker, SpaceQwen) remain in the low-to mid-40% range overall, indicating that specialization does not automatically produce higher MCQ accuracy across the full diversity of SPATIALAB. The overall pattern is favorable to instruction tuning and reasoning capability for geometric and orientation questions, while the persistent margins on navigation and multi-step grounding identify the remaining practical bottleneck.

SPATIALAB-OPEN. Table 3 reports open-ended accuracy under free-form generation and LLM-judge scoring. Scores are consistently lower than the corresponding MCQ results: overall model accuracy ranges from about 9.6% (InternVL3.5-1B) to 40.9% (GPT-5-mini), compared with a 64.9% human baseline. Proprietary models lead the open-ended leaderboard, including GPT-5-mini at 40.93%, Gemini-2.5-Flash at 30.93%, and GPT-4o-mini at 26.00%. Reasoning-tuned systems are also competitive, with o4-mini reaching 37.86% and Gemini-2.5-Flash-Thinking reaching 32.77%, showing that instruction- or CoT-style tuning improves generative spatial answers relative to many

baseline open-source models. Small open-source models mostly cluster in the teens (InternVL3.5-1B = 9.64%, Qwen-VL2.5-3B = 12.93%), while larger open models move into the low-20% range (InternVL3.5-72B = 23.36%, Qwen-VL2.5-72B = 24.64%). Spatial-specialist models remain around 10–13% overall, so task-specific training does not transfer robustly to open-ended generation. Subtask patterns mirror the MCQ evidence: top models are relatively stronger on orientation and some relative-position questions (e.g., GPT-5-mini Rel.Pos = 49.53%, Ori = 37.13%), whereas depth & occlusion, size & scale, and especially spatial navigation remain consistently difficult, with many scores below <30%.

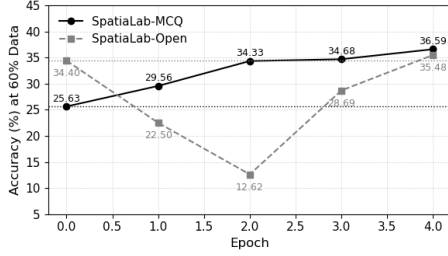
5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

Across 25 models, the average MCQ→open-ended gap in SPATIALAB is 23.0% ($\sigma = 5.5\%$), and this gap is stable across subtask means, ranging from 22.89% for spatial navigation to 24.57% for 3D geometry. Spatial-specialist models exhibit the largest average gaps (about 27%), with especially large drops in spatial navigation (up to 36.68%) and orientation (34.44%). Reasoning-oriented models show smaller gaps (around 19%) and lower subtask variance, suggesting that instruction tuning and CoT decoding improve format robustness even when they do not close the absolute human gap. Correlations further clarify which categories drive the aggregate disparity: spatial navigation nearly determines the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also contributing strongly. Negative or near-zero gaps, such as Llama-3.2-11B at -1.98 in depth & occlusion and o4-mini at -3.18 in relative position, show that MCQ distractor structure can occasionally understate or distort competence. These results make the practical implication clear: MCQ accuracy alone can overestimate usable spatial reasoning when free-form generation is required. We therefore recommend reporting MCQ and open-ended scores together, auditing distractor effects, using stepwise generation or instruction tuning when appropriate, and presenting per-subtask diagnostics rather than only aggregate scores. More details are in Appendix G and Table 18.

5.3 ERROR ANALYSIS

SPATIALAB-MCQ. The MCQ error analysis in Appendix E shows a consistent pattern of high local peaks paired with weak holistic spatial competence. Closed-source models reach strong maxima, such as 85.71% in *Stacking Orientation*, but fall to near-chance in *Relative Size Comparison*. Open-source scaling raises some ceilings, including 80.56% in *Corner/Angle Positioning*, yet does not prevent severe failures such as 2.0% in *Object Rotation*. Reasoning-augmented and spatial-specialist models contribute localized gains in tasks such as occlusion inference or navigation, but many physical-abstract tasks, including *Gravity Effects* and *Stability Prediction*, remain capped below 55%. The resulting error distributions highlight weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*). Stronger scores in *Obstacle Avoidance* suggest that some models can exploit local shortcuts, but the cross-subcategory margins indicate that robust planning is still missing. The empirical takeaway is that current VLMs can achieve impressive task-specific peaks while lacking stable encodings of orientation, physics, and compositional spatial logic.

SPATIALAB-OPEN. The open-ended error analysis in Appendix F shows the same peak-versus-consistency trade-off under a harder generation condition. Top closed-source systems reach the strongest subcategory results, for example GPT-5-mini at 58.14% in *Directional Relations*, but still collapse on tasks such as *Proximity Gradients*. Large open-source models can score well in *Depth & Occlusion*, including a 60.0% result, yet fall to 9.3% on *Proximity*, showing that scale improves selected ceilings without ensuring robustness. Smaller models retain isolated strengths, such as 26.0% in *Relative Size Comparison*, but can fail completely, as in 0.0% on *Betweenness*. Reasoning-tuned systems provide the most favorable evidence for explicit reasoning—Gemini-2.5-Flash-Thinking reaches 75.0% in *Tool Handedness*—but comparable systems still fall to single digits in other subcategories. Specialized spatial models rarely exceed 20% overall and frequently fail on core relational tasks. These quantitative patterns point to missing representations for spatial analogies, limits tied to scale and training diversity, and brittle reasoning pipelines; progress will require unified geometric encodings, physics-aware reasoning, and embodied data to turn isolated peaks into consistent spatial competence.



Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(a) SFT accuracy trends over epochs, showing how fine-tuning changes performance over training time. (b) Before-versus-after SFT accuracy by category for MCQ and open-ended formats, highlighting where gains transfer across evaluation conditions.

Figure 4: SFT training results: (a) accuracy trends over epochs, and (b) final before-versus-after accuracy values by format.

Qualitative Error Analysis. The qualitative analysis in Appendix I supports the quantitative findings by showing that errors cluster into recurring, interpretable classes rather than random noise. Common failures include spatial mislocalization in crowded scenes, perspective and scale mistakes caused by overreliance on object-size priors, and occlusion or ordering failures for thin or partially hidden structures. We also observe attribute confusion, where perceptual properties are mixed with functional ones, and open-ended rationalizations that are fluent but visually ungrounded. These patterns are consistent across architectures and prompting styles, which suggests structural biases rather than isolated dataset artifacts. Failure rates rise when multiple cues must be combined, such as depth ordering with relative size, and confidence calibration is especially weak in open-ended settings. Diagnostic tests show that models often ignore small but decisive visual features, rely on brittle heuristics, and fail to update under counterfactual edits. Together, the qualitative and quantitative evidence indicates that current VLMs can perform coarse perception but struggle with multi-cue integration, motivating geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. Reasoning-enabled models consistently improve over their corresponding baselines in both MCQ and open-ended formats, with the largest reported gain in relational and orientation tasks (e.g., +13.1% in Relative Positioning for MCQ). The gains are most favorable where the task can be stabilized by logical comparison, but they remain uneven across categories and include declines in Size & Scale. Thus, inherent reasoning improves some accuracy margins and stabilizes open-ended behavior, but it does not fully solve visual grounding or scale sensitivity. Detailed per-category comparisons are provided in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Table 19 shows that CoT prompting provides limited average benefit and often reduces accuracy, with Orientation as the clearest category showing consistent gains. This comparison suggests that step-by-step verbal reasoning helps when the spatial relation can be decomposed into directional alignment, but it does not reliably improve depth, scale, or navigation tasks that require stronger perceptual grounding. Unlike in many text-only settings, CoT on SPATIALAB can amplify flawed visual priors rather than correct them, so its practical value is category-dependent (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Adding self-reflection to CoT, where models are prompted to review and correct their prior reasoning step by step (see Figure 8 in Appendix), yields modest MCQ gains, especially for geometry and depth. However, these gains do not transfer reliably to open-ended evaluation (Appendix H.3). The format split is important: reflection can improve constrained selection, but the same mechanism does not compensate for missing perceptual evidence when models must generate grounded answers.

Supervised Fine-Tuning (SFT). As shown in Figure 4, supervised fine-tuning (SFT) of Qwen2.5-VL-3B-Instruct on a stratified 40% split of SPATIALAB (~560 samples) consistently improves MCQ accuracy across all spatial reasoning categories. This is one of the clearest intervention-level improvements in the paper. At the same time, transfer to open-ended tasks is limited and sometimes negative, indicating that the gain is concentrated in the structured answer condition. This divergence suggests overfitting to task-specific answer distributions and supports

concerns that SFT can *teach to the test* rather than produce generalizable reasoning (Wang et al., 2025). The open-ended drop points to internal representations tuned for categorical discrimination but unstable in generative settings, potentially due to catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a) and objectives that favor discrete-choice alignment. Overall, SFT improves constrained accuracy substantially, but the MCQ–open gap shows why both formats are necessary for evaluating practical spatial reasoning. More details are available in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). To test agentic decomposition, we use SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) for structured spatial reasoning on images. The framework decomposes visual reasoning into sub-tasks handled by agents for object segmentation, attribute extraction, spatial relation mapping, and transformation tracking, then consolidates those outputs into a shared structured representation. We implemented the pipeline with Gemini-2.5-Flash at low temperature to make the multi-stage refinement process deterministic. The strongest quantitative effect appears in Orientation, where performance improves by +8.00% in MCQ and +36.00% in open-ended evaluation. The same agentic setup, however, produces large declines in Depth & Occlusion (-24.00%) and Spatial Navigation (-12.00%) under open evaluation. This contrast is informative: agentic workflows can substantially improve tasks reducible to sequential alignment, but when the underlying perceptual priors for depth or occlusion are weak, multi-step reasoning can propagate errors instead of correcting them. More details are available in Appendix H.5.

6 DISCUSSION

SPATIALAB reframes spatial-reasoning evaluation as a set of explicit quantitative comparisons: 1,400 real-world visual QA pairs, six major categories, 30 subcategories, two answer formats, and human baselines. This structure matters because it separates option-based recognition from open-ended generative grounding and makes category-level margins visible. Across more than 25 contemporary VLMs, the benchmark shows that current systems can achieve localized strengths but remain far from human performance under realistic visual variability. The multi-stage review pipeline and complexity profiling over lighting, texture, object visibility, and spatial relations make the resulting gaps interpretable as model limitations rather than artifacts of a narrow evaluation format.

The main empirical pattern is the consistency of the gap. In MCQ evaluation, the best models remain near the mid-50% range while humans exceed 87%; in open-ended evaluation, models lose another 10–25% and the strongest result is 40.93% against a 64.93% human baseline. Error analysis shows that the shortfall is not uniform: transparency, occlusion, multi-step navigation, and 3D geometric relations are repeatedly harder than orientation or some relative-position tasks. Interventions also show favorable but bounded effects. SFT produces clear gains in structured MCQ tasks, yet can destabilize open-ended reasoning, suggesting format-specific optimization and possible catastrophic forgetting of linguistic priors. CoT and self-reflection help most when tasks reduce to sequential alignment, but they do not reliably repair depth or scale errors. Agentic decomposition similarly improves Orientation while degrading Depth & Occlusion and Spatial Navigation in open evaluation. These comparisons indicate that scaling, prompting, and multi-agent scaffolds are useful but insufficient on their own; closing the measured margins will require models with stronger geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

SPATIALAB provides a quantitative, real-world view of spatial reasoning in VLMs through 1,400 validated QA pairs spanning six spatial categories and thirty subcategories. The paired MCQ and open-ended formats make the main empirical pattern clear: current models achieve localized strengths but remain far below human baselines, with the best MCQ result at 54.93% versus 87.57% for humans and the best open-ended result at 40.93% versus 64.93% for humans. The additional 10–25% drop from MCQ to open-ended evaluation shows that answer format materially changes the estimate of practical competence. Error analyses further show that the largest and most consistent weaknesses occur in depth, occlusion, 3D geometry, and multi-step navigation, indicating reliance on surface cues rather than stable grounded spatial representations. SPATIALAB therefore offers both a benchmark and a diagnostic comparison framework for measuring progress. The intervention results reinforce

this conclusion. Reasoning, CoT, self-reflection, SFT, and agentic decomposition can improve selected categories or formats, but the gains are not uniformly reliable across the benchmark. Future progress will require richer training signals, such as embodied interaction and action-conditioned feedback, together with architectures that support geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We addressed ethical considerations throughout benchmark construction so that the reported quantitative comparisons do not depend on unnecessary personal or sensitive information. Image metadata was removed to protect privacy, and personally identifiable features, including human faces and license plates, were minimized or excluded wherever possible. The dataset was curated with attention to sourcing and licensing to reduce copyright risk, although minor oversights may occur despite these safeguards. The construction process was designed for transparency and fairness, with no demographic group singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To support reproducible comparison, we document the dataset construction process, annotation pipeline, evaluation protocols, filtering steps, quality-control procedures, and validation criteria in the main text and appendix. The task categories and annotation criteria are defined explicitly so that future work can interpret per-category and per-format metrics consistently. Hyperparameters, model configurations, and training procedures are reported in the appendix to support replication of the empirical results.

REFERENCES

- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. In *NeurIPS*, 2022.
- Anthropic. Model Card Addendum: Claude 3.5 Haiku and Upgraded Claude 3.5 Sonnet, October 2024. URL <https://assets.anthropic.com/m/1cd9d098ac3e6467/original/Claude-3-Model-Card-October-Addendum.pdf>. Accessed: 2025-09-11.
- Daichi Azuma, Taiki Miyanishi, Shuhei Kurita, and Motoaki Kawanabe. Scanqa: 3d question answering for spatial scene understanding. In *proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 19129–19139, 2022.
- Alan Baddeley. Working memory. *Comptes Rendus de l’Académie des Sciences-Series III-Sciences de la Vie*, 321(2-3):167–173, 1998.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. URL <https://arxiv.org/abs/2502.13923>.
- Yoav Bar-Anan, Nira Liberman, and Yaacov Trope. The association between psychological distance and construal level: evidence from an implicit association test. *Journal of experimental psychology: General*, 135(4):609, 2006.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian

- 594 Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil J. Joshi, Ryan Julian, Dmitry Kalash-
595 nikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha
596 Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl
597 Pertsch, Jornell Quiambao, Kanishka Rao, Michael S. Ryoo, Grecia Salazar, Pannag R. Sanketi,
598 Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong T. Tran, Vincent
599 Vanhoucke, Steve Vega, Quan Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and
600 Brianna Zitkovich. RT-1: robotics transformer for real-world control at scale. In *Robotics: Science
601 and Systems XIX, Daegu, Republic of Korea, July 10-14, 2023*, 2023.
- 602 Wenxiao Cai, Yaroslav Ponomarenko, Jianhao Yuan, Xiaoqi Li, Wankou Yang, Hao Dong, and
603 Bo Zhao. Spatialbot: Precise spatial understanding with vision language models. *CoRR*,
604 abs/2406.13642, 2024.
- 605 Zhongang Cai, Yubo Wang, Qingping Sun, Ruisi Wang, Chenyang Gu, Wanqi Yin, Zhiqian Lin,
606 Zhitao Yang, Chen Wei, Xuanke Shi, Kewang Deng, Xiaoyang Han, Zukai Chen, Jiaqi Li, Xiangyu
607 Fan, Hanming Deng, Lewei Lu, Bo Li, Ziwei Liu, Quan Wang, Dahua Lin, and Lei Yang. Has
608 gpt-5 achieved spatial intelligence? an empirical study, 2025. URL [https://arxiv.org/
609 abs/2508.13142](https://arxiv.org/abs/2508.13142).
- 610 Christopher F Chabris, Thomas E Jerde, Anita W Woolley, Margaret E Gerbasi, Jonathon P Schuldt,
611 Sean L Bennett, J Richard Hackman, and Stephen M Kosslyn. Spatial and object visualization
612 cognitive styles: Validation studies in 3800 individuals. *Group brain technical report*, 2(1-20):2,
613 2006.
- 614 Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Danny Driess, Pete Florence, Dorsa Sadigh,
615 Leonidas Guibas, and Fei Xia. Spatialvlm: Endowing vision-language models with spatial
616 reasoning capabilities. *arXiv preprint arXiv:2401.12168*, 2024a. URL [https://arxiv.org/
617 abs/2401.12168](https://arxiv.org/abs/2401.12168).
- 618 Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Dorsa Sadigh, Leonidas J. Guibas, and Fei Xia.
619 Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. In *IEEE/CVF
620 Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June
621 16-22, 2024*, pp. 14455–14465. IEEE, 2024b.
- 622 Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor
623 Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for vlms? an attention
624 mechanism perspective on focus areas. *CoRR*, abs/2503.01773, 2025.
- 625 An-Chieh Cheng, Hongxu Yin, Yang Fu, Qiushan Guo, Ruihan Yang, Jan Kautz, Xiaolong Wang, and
626 Sifei Liu. Spatialrgpt: Grounded spatial reasoning in vision-language models. In Amir Globersons,
627 Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng
628 Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on
629 Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December
630 10 - 15, 2024*, 2024.
- 631 Open X.-Embodiment Collaboration, Abhishek Padalkar, Acorn Pooley, Ajinkya Jain, Alex Bewley,
632 Alexander Herzog, Alex Irpan, Alexander Khazatsky, Anant Raj, Anikait Singh, Anthony Brohan,
633 Antonin Raffin, Ayzaan Wahid, Ben Burgess-Limerick, Beomjoon Kim, Bernhard Schölkopf,
634 Brian Ichter, Cewu Lu, Charles Xu, Chelsea Finn, Chenfeng Xu, Cheng Chi, Chenguang Huang,
635 Christine Chan, Chuer Pan, Chuyuan Fu, Coline Devin, Danny Driess, Deepak Pathak, Dhruv Shah,
636 Dieter Büchler, Dmitry Kalashnikov, Dorsa Sadigh, Edward Johns, Federico Ceola, Fei Xia, Freek
637 Stulp, Gaoyue Zhou, Gaurav S. Sukhatme, Gautam Salhotra, Ge Yan, Giulio Schiavi, Gregory
638 Kahn, Hao Su, Haoshu Fang, Haochen Shi, Heni Ben Amor, Henrik I. Christensen, Hiroki Furuta,
639 Homer Walke, Hongjie Fang, Igor Mordatch, Ilija Radosavovic, and et al. Open x-embodiment:
640 Robotic learning datasets and RT-X models. *CoRR*, abs/2310.08864, 2023.
- 641 Lee J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334,
642 September 1951. ISSN 1860-0980. doi: 10.1007/bf02310555. URL [http://dx.doi.org/
643 10.1007/BF02310555](http://dx.doi.org/10.1007/BF02310555).
- 644 Angela Dai, Angel X. Chang, Manolis Savva, Maciej Halber, Thomas A. Funkhouser, and Matthias
645 Nießner. Scannet: Richly-annotated 3d reconstructions of indoor scenes. In *2017 IEEE Conference
646*

on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017, pp. 2432–2443. IEEE Computer Society, 2017.

Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch, and Pete Florence. Palm-e: An embodied multimodal language model. In *ICML*, 2023.

Chris Drummond. Replicability is not reproducibility: nor is it good science. In *Proceedings of the Evaluation Methods for Machine Learning Workshop at the 26th ICML*, volume 1, Montreal, Canada, 2009. National Research Council of Canada.

Chris Drummond and Nathalie Japkowicz. Warning: statistical benchmarking is addictive. kicking the habit in machine learning. *Journal of Experimental and Theoretical Artificial Intelligence*, 22(1):67–80, March 2010. ISSN 1362-3079. doi: 10.1080/09528130903010295. URL <http://dx.doi.org/10.1080/09528130903010295>.

Mengfei Du, Binhao Wu, Zejun Li, Xuanjing Huang, and Zhongyu Wei. Embspatial-bench: Benchmarking spatial understanding for embodied tasks with large vision-language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, ACL 2024 - Short Papers, Bangkok, Thailand, August 11-16, 2024*, pp. 346–355. Association for Computational Linguistics, 2024.

Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: multimodal large language models can see but not perceive. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol (eds.), *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part XXIII*, volume 15081 of *Lecture Notes in Computer Science*, pp. 148–166. Springer, 2024.

Howard E Gardner. *Frames of mind: The theory of multiple intelligences*. Basic books, 2011.

Gemma Team, Google DeepMind. Gemma 3 technical report, 2025. URL <https://storage.googleapis.com/deepmind-media/gemma/Gemma3Report.pdf>. Accessed: 2025-09-11.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne,

702 Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal
 703 Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong,
 704 Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic,
 705 Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie
 706 Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana
 707 Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie,
 708 Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon
 709 Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collet, Suchin Gururangan,
 710 Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas
 711 Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami,
 712 Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti,
 713 Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier
 714 Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao
 715 Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song,
 716 Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe
 717 Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya
 718 Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei
 719 Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu,
 720 Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit
 721 Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury,
 722 Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer,
 723 Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu,
 724 Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido,
 725 Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changan Wang, Changkyu
 726 Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer,
 727 Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu,
 728 Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc
 729 Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily
 730 Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers,
 731 Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank
 732 Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee,
 733 Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan,
 734 Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph,
 735 Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog,
 736 Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James
 737 Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny
 738 Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings,
 739 Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai
 740 Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik
 741 Veeraraghavan, Kelly Michelen, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle
 742 Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng
 743 Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madihan Khabsa, Manav Avalani, Manish
 744 Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim
 745 Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle
 746 Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang,
 747 Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam,
 748 Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier,
 749 Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia
 750 Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro
 751 Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani,
 752 Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy,
 753 Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin
 754 Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Sibi, Sai Jayesh Bondu,
 755 Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh
 Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay,
 Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang,
 Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie
 Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta,
 Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman,

Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.

Tanmay Gupta and Aniruddha Kembhavi. Visual programming: Compositional visual reasoning without training. In *CVPR*, 2023.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, September 2020. doi: 10.1038/s41586-020-2649-2. URL <https://doi.org/10.1038/s41586-020-2649-2>.

Md Tanzib Hosain, Salman Rahman, Md Kishor Morol, and Md Rizwan Parvez. Xolver: Multi-agent reasoning with holistic experience learning just like an olympiad team, 2025. URL <https://arxiv.org/abs/2506.14234>.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.

Jianheng Huang, Leyang Cui, Ante Wang, Chengyi Yang, Xinting Liao, Linfeng Song, Junfeng Yao, and Jinsong Su. Mitigating catastrophic forgetting in large language models with self-synthesized rehearsal. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1416–1428, Bangkok, Thailand, August 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.77. URL <https://aclanthology.org/2024.acl-long.77/>.

Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner monologue: Embodied reasoning through planning with language models. In *CoRL*, 2022.

Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *CoRL*, 2024b.

Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*, pp. 6700–6709. Computer Vision Foundation / IEEE, 2019.

Mengdi Jia, Zekun Qi, Shaochen Zhang, Wen Yao Zhang, Xinqiang Yu, Jiawei He, He Wang, and Li Yi. Omnispatial: Towards comprehensive spatial reasoning benchmark for vision language models, 2025. URL <https://arxiv.org/abs/2506.03135>.

Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick. Clevr: A diagnostic dataset for compositional language and elementary visual reasoning, 2016. URL <https://arxiv.org/abs/1612.06890>.

Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? investigating their struggle with spatial reasoning. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023*, pp. 9161–9175. Association for Computational Linguistics, 2023.

810 Terry K. Koo and Mae Y. Li. A guideline of selecting and reporting intraclass correlation coefficients
811 for reliability research. *Journal of Chiropractic Medicine*, 15(2):155–163, June 2016. ISSN
812 1556-3707. doi: 10.1016/j.jcm.2016.02.012. URL [http://dx.doi.org/10.1016/j.jcm.](http://dx.doi.org/10.1016/j.jcm.2016.02.012)
813 2016.02.012.

814 Eliza Kosoy, Annya Dahmani, Andrew K. Lampinen, Iulia M. Comsa, Soojin Jeong, Ishita Dasgupta,
815 and Kelsey Allen. Decoupling the components of geometric understanding in vision language
816 models. *CoRR*, abs/2503.03840, 2025.

817 Junnan Li, Dongxu Li, Silvio Savarese, and Steven C. H. Hoi. BLIP-2: bootstrapping language-image
818 pre-training with frozen image encoders and large language models. *CoRR*, abs/2301.12597, 2023.

819 Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. *Trans. Assoc. Comput.*
820 *Linguistics*, 11:635–651, 2023.

821 Siqi Liu, Ian Gemp, Luke Marris, Georgios Piliouras, Nicolas Heess, and Marc Lanctot. Re-
822 evaluating open-ended evaluation of large language models. In *The Thirteenth International*
823 *Conference on Learning Representations*, 2025. URL [https://openreview.net/forum?](https://openreview.net/forum?id=kbOAIXKWgx)
824 [id=kbOAIXKWgx](https://openreview.net/forum?id=kbOAIXKWgx).

825 Yao Mu, Qinglong Zhang, Mengkang Hu, Wenhai Wang, Mingyu Ding, Jun Jin, Bin Wang, Jifeng
826 Dai, Yu Qiao, and Ping Luo. Embodiedgpt: Vision-language pre-training via embodied chain of
827 thought. In *NeurIPS*, 2023.

828 OpenAI. Gpt-4v(ision) system card, 2023. URL [https://openai.com/research/](https://openai.com/research/gpt-4v-system-card)
829 [gpt-4v-system-card](https://openai.com/research/gpt-4v-system-card).

830 OpenAI. Gpt-4o system card, 2024. URL <https://arxiv.org/abs/2410.21276>.

831 OpenAI. Openai o3 and o4-mini system card, April 2025. URL [https://openai.com/index/](https://openai.com/index/o3-o4-mini-system-card/)
832 [o3-o4-mini-system-card/](https://openai.com/index/o3-o4-mini-system-card/). System Cards. Accessed: 2025-09-11.

833 Stephen Palmer. *Vision Science: From Photons to Phenomenology*, volume 1. MIT Press, 01 1999.

834 Atin Pothiraj, Elias Stengel-Eskin, Jaemin Cho, and Mohit Bansal. Capture: Evaluating spatial
835 reasoning in vision language models via occluded object counting. *CoRR*, abs/2504.15485, 2025.

836 Fred H Previc. The neuropsychology of 3-d space. *Psychological bulletin*, 124(2):123, 1998.

837 Zekun Qi, Runpei Dong, Shaochen Zhang, Haoran Geng, Chunrui Han, Zheng Ge, Li Yi, and
838 Kaisheng Ma. Shapellm: Universal 3d object understanding for embodied interaction. In *Computer*
839 *Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024,*
840 *Proceedings, Part XLIII*, volume 15101 of *Lecture Notes in Computer Science*, pp. 214–238.
841 Springer, 2024.

842 Zekun Qi, Wenyao Zhang, Yufei Ding, Runpei Dong, Xinqiang Yu, Jingwen Li, Lingyun Xu, Baoyu
843 Li, Xialin He, Guofan Fan, Jiazhaio Zhang, Jiawei He, Jiayuan Gu, Xin Jin, Kaisheng Ma, Zhizheng
844 Zhang, He Wang, and Li Yi. Sofar: Language-grounded orientation bridges spatial reasoning and
845 object manipulation. *CoRR*, abs/2502.13143, 2025.

846 Tiago Ramalho, Tomás Kociský, Frederic Besse, S. M. Ali Eslami, Gábor Melis, Fabio Viola, Phil
847 Blunsom, and Karl Moritz Hermann. Encoding spatial relations from natural language. *CoRR*,
848 abs/1807.01670, 2018.

849 Fatemeh Shiri, Xiao-Yu Guo, Mona Far, Xin Yu, Reza Haf, and Yuan-Fang Li. An empirical analysis
850 on spatial reasoning capabilities of large multimodal models. In Yaser Al-Onaizan, Mohit Bansal,
851 and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural*
852 *Language Processing, EMNLP 2024, Miami, FL, USA, November 12-16, 2024*, pp. 21440–21455.
853 Association for Computational Linguistics, 2024.

854 Patrick E. Shrout and Joseph L. Fleiss. Intraclass correlations: Uses in assessing rater reliability.
855 *Psychological Bulletin*, 86(2):420–428, 1979. ISSN 0033-2909. doi: 10.1037/0033-2909.86.2.420.
856 URL <http://dx.doi.org/10.1037/0033-2909.86.2.420>.

- Chan Hee Song, Valts Blukis, Jonathan Tremblay, Stephen Tyree, Yu Su, and Stan Birchfield. Robospatal: Teaching spatial understanding to 2d and 3d vision-language models for robotics. *CoRR*, abs/2411.16537, 2024.
- StepFun. Step-3 is large yet affordable: Model-system co-design for cost-effective decoding, 2025. URL <https://arxiv.org/abs/2507.19427>.
- Ilias Stogiannidis, Steven McDonagh, and Sotirios A. Tsaftaris. Mind the gap: Benchmarking spatial reasoning in vision-language models. *CoRR*, abs/2503.19707, 2025.
- Emilia Szymanska, Mihai Dusmanu, Jan-Willem Buurlage, Mahdi Rad, and Marc Pollefeys. Space3d-bench: Spatial 3d question answering benchmark. *CoRR*, abs/2408.16662, 2024.
- Mohsen Tavakol and Reg Dennick. Making sense of cronbach’s alpha. *International Journal of Medical Education*, 2:53–55, June 2011. ISSN 2042-6372. doi: 10.5116/ijme.4dfb.8dfd. URL <http://dx.doi.org/10.5116/ijme.4dfb.8dfd>.
- Gemini Team. Gemini: A family of highly capable multimodal models, 2025a. URL <https://arxiv.org/abs/2312.11805>.
- Kimi Team. Kimi-vl technical report, 2025b. URL <https://arxiv.org/abs/2504.07491>.
- V Team, Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, Lihang Pan, Shuaiqi Duan, Weihang Wang, Yan Wang, Yean Cheng, Zehai He, Zhe Su, Zhen Yang, Ziyang Pan, Aohan Zeng, Baoxu Wang, Bin Chen, Boyan Shi, Changyu Pang, Chenhui Zhang, Da Yin, Fan Yang, Guoqing Chen, Jiazheng Xu, Jiale Zhu, Jiali Chen, Jing Chen, Jinhao Chen, Jinghao Lin, Jinjiang Wang, Junjie Chen, Leqi Lei, Letian Gong, Leyi Pan, Mingdao Liu, Mingde Xu, Mingzhi Zhang, Qinkai Zheng, Sheng Yang, Shi Zhong, Shiyu Huang, Shuyuan Zhao, Siyan Xue, Shangqin Tu, Shengbiao Meng, Tianshu Zhang, Tianwei Luo, Tianxiang Hao, Tianyu Tong, Wenkai Li, Wei Jia, Xiao Liu, Xiaohan Zhang, Xin Lyu, Xinyue Fan, Xuancheng Huang, Yanling Wang, Yadong Xue, Yanfeng Wang, Yanzi Wang, Yifan An, Yifan Du, Yiming Shi, Yiheng Huang, Yilin Niu, Yuan Wang, Yuanchang Yue, Yuchen Li, Yutao Zhang, Yuting Wang, Yu Wang, Yuxuan Zhang, Zhao Xue, Zhenyu Hou, Zhengxiao Du, Zihan Wang, Peng Zhang, Debing Liu, Bin Xu, Juanzi Li, Minlie Huang, Yuxiao Dong, and Jie Tang. Glm-4.5v and glm-4.1v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning, 2025. URL <https://arxiv.org/abs/2507.01006>.
- Yaacov Trope and Nira Liberman. Construal-level theory of psychological distance. *Psychological review*, 117(2):440, 2010.
- Raphael Vallat. Pingouin: statistics in python. *Journal of Open Source Software*, 3(31):1026, November 2018. doi: 10.21105/joss.01026.
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Yubo Wang, Xiang Yue, and Wenhui Chen. Critique fine-tuning: Learning to critique is more effective than learning to imitate, 2025. URL <https://arxiv.org/abs/2501.17703>.
- Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in space: How multimodal large language models see, remember, and recall spaces. *CoRR*, abs/2412.14171, 2024.
- Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Ehsan Azarnasab, Faisal Ahmed, Zicheng Liu, Ce Liu, Michael Zeng, and Lijuan Wang. MM-REACT: prompting chatgpt for multimodal reasoning and action. *CoRR*, abs/2303.11381, 2023.

-
- Wentao Yuan, Jiafei Duan, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithyavairavan Murali, Arsalan Mousavian, and Dieter Fox. Robopoint: A vision-language model for spatial affordance prediction for robotics. *CoRR*, abs/2406.10721, 2024.
- Kaizhi Zheng, Xiaotong Chen, Odest Chadwicke Jenkins, and Xin Eric Wang. Vlmbench: A compositional benchmark for vision-and-language manipulation, 2022. URL <https://arxiv.org/abs/2206.08522>.
- Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Xuehui Wang, Yue Cao, Yangzhou Liu, Xingguang Wei, Hongjie Zhang, Haomin Wang, Weiye Xu, Hao Li, Jiahao Wang, Nianchen Deng, Songze Li, Yinan He, Tan Jiang, Jiapeng Luo, Yi Wang, Conghui He, Botian Shi, Xingcheng Zhang, Wenqi Shao, Junjun He, Yingdong Xiong, Wenwen Qu, Peng Sun, Penglong Jiao, Han Lv, Lijun Wu, Kaipeng Zhang, Huipeng Deng, Jiaye Ge, Kai Chen, Limin Wang, Min Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai, and Wenhai Wang. Internv13: Exploring advanced training and test-time recipes for open-source multimodal models, 2025. URL <https://arxiv.org/abs/2504.10479>.
- Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong T. Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning, CoRL 2023, 6-9 November 2023, Atlanta, GA, USA*, volume 229 of *Proceedings of Machine Learning Research*, pp. 2165–2183. PMLR, 2023.

972	A Related Work	21
973	A.1 Detailed Analysis of Limitations in Prior Benchmarks	22
974		
975	B Task Design and Descriptions	22
976	B.1 Relative Positioning	23
977	B.2 Depth & Occlusion	24
978	B.3 Orientation	24
979	B.4 Size & Scale	24
980	B.5 Spatial Navigation	25
981	B.6 3D Geometry	25
982		
983	C More Details on Benchmark Construction	25
984	C.1 Annotator Recruitment	25
985	C.2 Annotator Training and Protocol	26
986		
987	D More details on Experiments and Evaluation	26
988	D.1 Multiple Choice Evaluation	26
989	D.2 Open-Ended Evaluation	26
990	D.2.1 Evaluation of LLM-as-a-Judge for Open-ended Evaluation	26
991		
992	E Error Analysis of SPATIALAB-MCQ	28
993	E.1 Sub-Category-wise Quantitative Error Analysis	28
994	E.2 Model Failure Analysis	31
995		
996	F Error Analysis of SPATIALAB-OPEN	33
997	F.1 Sub-Category-wise Quantitative Error Analysis	33
998	F.2 Model Failure Analysis	35
999		
1000	G Analysis of MCQ vs. Open-Ended Performance Gaps	36
1001	G.1 Quantitative snapshot	36
1002	G.2 Patterns and observations	37
1003	G.3 Hypotheses on root causes	38
1004	G.4 Practical implications	39
1005		
1006	H Details on Improving Visual Reasoning Capabilities	39
1007	H.1 Inherent Reasoning Mechanisms	39
1008	H.2 Chain-of-Thought (CoT) Prompting	40
1009	H.3 Chain-of-Thought (CoT) Prompting with Self-Reflection	41
1010	H.4 Supervised Fine-Tuning (SFT)	42
1011	H.4.1 Setup and Implementation	42
1012	H.4.2 Supervised Fine-Tuning (SFT) Performance	42
1013		
1014		
1015		
1016		
1017		
1018		
1019		
1020		
1021		
1022		
1023		
1024		
1025		

1026	H.4.3 SFT Dynamics Analysis	44
1027	H.4.4 Performance Gain on External Benchmarks	45
1028		
1029	H.5 AI Agents for Spatial Reasoning	46
1030	H.5.1 Agents	46
1031	H.5.2 Technical Implementation Details	47
1032	H.5.3 Evaluation and Analysis	47
1033		
1034		
1035		
1036	I Qualitative Error Analysis and Error Patterns	48
1037	I.1 Overview	48
1038	I.2 Error taxonomy and analysis	48
1039	I.2.1 Spatial mislocalization and reference confusion.	48
1040	I.2.2 Perspective and scale mistakes.	49
1041	I.2.3 Occlusion and ordering failures.	49
1042	I.2.4 Attribute confusion and semantic swap.	50
1043	I.2.5 Open-ended rationalization without verification.	50
1044	I.3 Error patterns and cross-model signals	51
1045	I.4 Mitigation strategies	51
1046		
1047		
1048		
1049		
1050		
1051	J Representative Benchmark Samples Across All Sub-Categories	56
1052		
1053	K Statistical Robustness and Dataset Stability	86
1054	K.1 Model Robustness	86
1055	K.1.1 Multiple Run Averages and Deviations	86
1056	K.1.2 Intra-Class Correlation (ICC)	86
1057	K.2 Dataset Stability and Internal Consistency	86
1058	K.2.1 Resampling Study	87
1059	K.2.2 Item-Level Consistency	87
1060	K.2.3 Cronbach’s Alpha	87
1061	K.3 Empirical Results	88
1062	K.3.1 Model Evaluation Robustness	88
1063	K.3.2 Dataset Stability and Internal Consistency	89
1064	K.3.3 Overall Discussion	90
1065		
1066		
1067		
1068		
1069		
1070		
1071		
1072		
1073		
1074		
1075		
1076		
1077		
1078		
1079		

A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning is commonly defined as the ability to understand geometric structure and relational layout in visual scenes, and it is central to embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) enabled controlled measurement of compositional reasoning through synthetic or curated templates, but their ecological validity was limited. With large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023), the field has moved toward more naturalistic evaluation, yet the empirical pattern remains uneven: strong captioning or visual-QA performance does not reliably transfer to depth ordering, occlusion inference, or reference-frame alignment. This gap motivates benchmarks that report not only overall scores but also category-specific margins under realistic visual conditions.

Spatial reasoning benchmarks. Recent benchmarks probe these weaknesses from different angles (Table 1). SCANQA (Azuma et al., 2022) introduced embodied QA over 3D indoor environments, but its scale (800 scenes) and relatively shallow complexity constrain generalization analysis. VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) broadened evaluation to real-world or household contexts, but template-driven questions reduce linguistic variability and emphasize simple binary or multiple-choice relations. More recent datasets, including EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), increase task variety but remain limited by scale (2.2K and 211 samples, respectively) and annotation style. SPATIALRGPT-BENCH and BLINK-SPATIAL add mixed domains and up to 14 categories, improving coverage while still emphasizing static snapshots or limited multi-view simulation.

Embodiment, difficulty, and diversity. Embodied datasets provide another comparison axis. ROBOSPATIAL (Song et al., 2024) scales to 1M scenes with robot-centric questions, but template-based annotation can make the distribution vulnerable to heuristic shortcuts. SPATIALVLM (Chen et al., 2024b) provides very large scale through billions of QA pairs, yet its two-category coverage limits fine-grained diagnosis. High-quality manual resources such as VSI-BENCH (Yang et al., 2024) improve annotation fidelity, but the 288-scene scale is too small for robust subcategory-level comparison. In complexity terms, many prior benchmarks remain around “medium” (Mm in Table 1) and rarely emphasize the hard or extremely hard cases that require multi-step causal or geometric inference. This leaves an evaluation gap for measuring next-generation VLMs under high-difficulty, real-world spatial conditions.

Evaluation protocols. Evaluation format strongly affects the meaning of reported scores. Closed-ended QA and binary judgments, common in benchmarks such as VISUAL SPATIAL and EMBSPATIAL-BENCH, simplify scoring but can inflate performance through option elimination. Open-ended evaluation, used in datasets such as ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN, provides a stronger test of generative grounding. SPATIALAB-OPEN pairs this format with high-complexity manually authored items and detailed error analysis, enabling comparisons beyond aggregate accuracy, including calibration behavior and cross-model divergence. Without this granularity, apparent gains can reflect dataset-specific shortcuts rather than durable spatial competence.

Positioning of SPATIALAB. SPATIALAB addresses these limitations by combining scale, taxonomy, format diversity, and human baselines in a single comparison framework. Compared with prior datasets spanning 2 to 14 task categories, SPATIALAB covers 30 subcategories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation. Unlike template-based benchmarks, its questions are manually authored and puzzle-inspired, increasing linguistic diversity and reducing shortcut opportunities. As shown in Table 1, even strong baselines such as InternVL3.5-72B and GPT-5-mini remain well below 60%, and the best open-ended score is 40.93%. These margins demonstrate that the benchmark is difficult enough to distinguish model families while still providing interpretable category-level evidence.

Toward next-generation evaluation. By combining high difficulty, broad coverage, manual annotation, paired MCQ/open-ended formats, and detailed diagnostics, SPATIALAB extends spatial-reasoning evaluation from narrow template tests toward comprehensive embodied measurement. The benchmark supports both correctness analysis and error-pattern analysis, including causal and

counterfactual reasoning failures and calibration reliability. It complements resources such as MIND THE GAP and VLM4D, which explore temporal or generative aspects but remain narrower in coverage. We expect SPATIALAB to serve as a high-bar diagnostic tool for tracking whether future VLMs improve aggregate scores, reduce category-specific margins, and move from surface-level recognition toward grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To situate SPATIALAB within the broader landscape, we examine prior spatial-reasoning benchmarks along the comparison dimensions most relevant to quantitative interpretation: domain, scale, task taxonomy, and evaluation format. Earlier efforts provide important foundations, but their structural constraints limit their ability to diagnose current VLMs under realistic spatial demands. The main limitations fall into three groups—synthetic-domain bias, limited task scope, and restrictive evaluation formats—as summarized in Table 4.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA use synthetic environments or template-generated language. These controlled conditions isolate compositional reasoning, but the visual domain lacks real-world complexity and the language distribution is narrow. As a result, high scores can reflect template regularities rather than grounded spatial interpretation. Simulated benchmarks such as ROBOSPATIAL face a related sim-to-real gap.

Domain and Scale Limitations. Datasets based on 3D reconstruction, including ScanQA and Space3D-Bench, provide valuable depth supervision but are largely restricted to indoor scenes. This limits evaluation in outdoor and mixed environments that matter for embodied agents. High-quality manual datasets such as VSI-Bench improve annotation quality but remain too small for stable high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM cover broad tasks but lack the detailed taxonomy needed to identify failures in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, rely heavily on Binary or MCQ formats, which can raise accuracy through option elimination rather than grounded generation. SPATIALAB mitigates these issues by pairing MCQ and Open-Ended evaluation across 30 spatial subcategories, making both aggregate and per-skill margins visible.

Table 4: Comparison-focused summary of limitations in representative spatial benchmarks. The table highlights how domain, evaluation format, and taxonomy affect the quantitative evidence each benchmark can provide relative to SPATIALAB.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale (< 1K scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Spatial reasoning requires more than object recognition; it depends on measurable judgments about how objects relate, align, occlude, and interact in physical space. *SpatialLab* operationalizes this ability through a balanced taxonomy whose categories and subcategories support fine-grained quantitative comparison. The tasks mirror everyday spatial questions, from relative positioning (e.g., *What’s to the left of the microwave?*) to geometric inference (e.g., *Would this stack fall if another book is added?*). Each category is decomposed into subcategories that isolate evidence about occlusion, orientation, navigation, and physical effects such as gravity. This structure lets us compare whether models merely detect objects or consistently perform grounded spatial interpretation across settings relevant to robotics, AR/VR, autonomous driving, and human-computer interaction. The following sections define each category and subcategory used in the benchmark.

The following task descriptions specify the comparison target for each category and the practical skill measured by each subcategory:

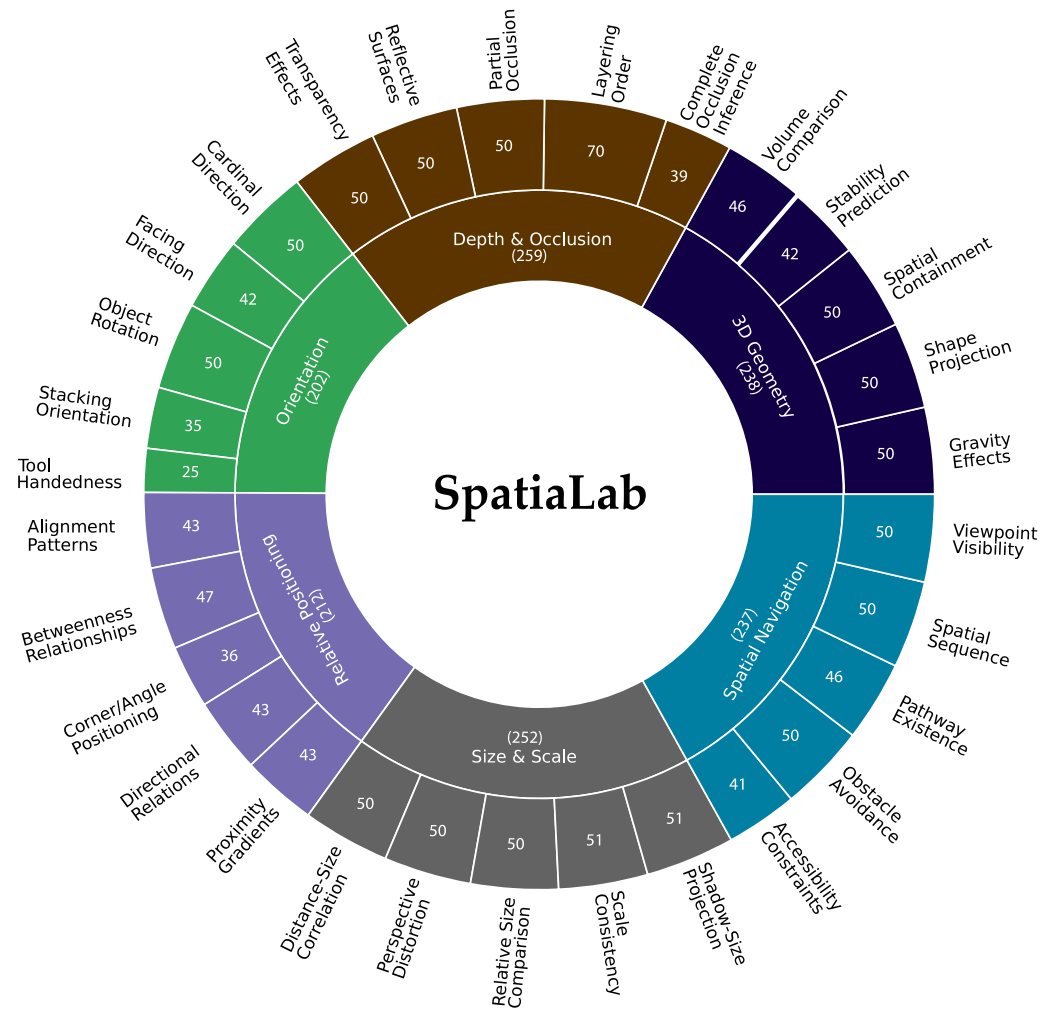


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each of the six categories decomposes into five subcategories, yielding thirty task types for balanced per-skill comparison.

B.1 RELATIVE POSITIONING

Directional Relations. Directional relations describe how objects are positioned relative to one another, such as left, right, above, or below. Understanding these relations enables models to interpret spatial layouts more naturally, just like how humans describe scenes. For example, identifying what lies immediately to the left of an appliance helps in scene understanding and robot navigation.

Proximity Gradients. Proximity reasoning involves recognizing which objects are closest or farthest from a reference point. This is key for questions like “Which object is nearest to the doorway?” Models capable of estimating gradients of closeness can support navigation tasks and assistive technologies.

Alignment Patterns. Alignment concerns whether objects follow a certain orientation or order, such as being stacked vertically or arranged in a straight line. Recognizing alignment patterns helps in understanding organization in environments like shelves, tables, or warehouses. Such reasoning is also critical for tasks like robotic sorting and automated inspection.

Betweenness Relationships. Betweenness focuses on identifying objects that occupy the middle ground between two references. For instance, asking what sits between the lamp and the armchair

requires reasoning about spatial continuity. This helps in navigation, retrieval, and situational awareness tasks.

Corner/Angle Positioning. Corners and angled positions often serve as reference anchors in spatial reasoning. Questions such as “What’s positioned in the back-right corner?” require the model to localize objects within constrained regions. This kind of reasoning is vital in robotics for mapping and placement.

B.2 DEPTH & OCCLUSION

Layering Order. Layering requires distinguishing which objects appear in front versus behind. For example, identifying the frontmost item on a table reflects depth reasoning. This is particularly useful in AR/VR applications where visual stacking affects user experience.

Partial Occlusion. Partial occlusion occurs when one object hides part of another from view. Recognizing the hidden item behind a coffee cup, for instance, requires filling in missing information. This strengthens robustness in perception systems operating in cluttered environments.

Complete Occlusion Inference. Sometimes objects are fully hidden, yet clues like shadows or bulges reveal their presence. Inferring what lies under a blanket requires the model to use indirect evidence. Such reasoning mirrors human inference in uncertain visual conditions.

Transparency Effects. Transparent surfaces, such as glass or water, allow partial visibility of objects behind them. Understanding what’s visible through a fish tank requires blending occlusion reasoning with transparency perception. This plays an important role in simulation, graphics, and embodied AI.

Reflective Surfaces. Reflective reasoning asks models to interpret mirrors, shiny metals, or glass reflections. Identifying what is reflected in a mirror involves indirect spatial mapping. This is essential for safety in autonomous vehicles and situational awareness in robotics.

B.3 ORIENTATION

Cardinal Direction. Cardinal reasoning relates to compass directions like north, south, east, and west. Asking which way a front door faces requires integrating global orientation cues. This is critical for navigation and geospatial alignment.

Object Rotation. Rotation reasoning measures how much an object has been turned around an axis. Estimating that a chair is rotated by some degrees is crucial for accurate 3D modeling. This helps robots align objects correctly for manipulation.

Facing Direction. Facing direction specifies the orientation of an object’s front side. For example, determining whether a monitor faces north requires mapping its surface orientation to global references. This is important in surveillance and robotics.

Stacking Orientation. Stacking involves understanding whether objects are piled vertically, horizontally, or in mixed orientations. Recognizing such layouts ensures stability in warehouses and homes. It is also critical for safe robotic stacking.

Tool Handedness. Certain tools, like scissors or gloves, are designed for left- or right-handed use. Identifying handedness requires fine-grained shape and orientation perception. This enhances assistive robotics in human-centered environments.

B.4 SIZE & SCALE

Relative Size Comparison. Relative comparisons assess whether one object is larger, taller, or smaller than another. Asking if a vase is taller than a statue requires proportional reasoning. This is crucial for packing, fitting, and recognition tasks.

Perspective Distortion. Perspective can make distant objects appear smaller than nearby ones, even if the actual size differs. Identifying illusions like a faraway car looking smaller than a nearby bicycle requires compensating for depth cues. This ability makes models more robust to camera perspectives.

1296 **Distance-Size Correlation.** Here, size judgments depend on distance estimation. For instance, a
1297 nearby bicycle appearing larger than a far car should not confuse true size reasoning. This helps in
1298 navigation, AR/VR, and visual grounding.

1299 **Scale Consistency.** Scale reasoning involves checking whether object sizes are reasonable in context.
1300 A mouse larger than a chair signals inconsistency, which models must detect. This strengthens realism
1301 in simulated environments.

1302 **Shadow-Size Projection.** Shadows provide indirect cues about object size and light source position.
1303 Estimating how tall a light must be based on shadow length requires geometric reasoning. This is
1304 useful in graphics, architecture, and physics-based modeling.

1305

1306 **B.5 SPATIAL NAVIGATION**

1307

1308 **Pathway Existence.** Pathway reasoning checks if a clear route exists between two points. For
1309 example, determining if one can reach the desk without moving chairs tests navigability. This ability
1310 underpins indoor robot planning.

1311 **Obstacle Avoidance.** Avoidance requires finding routes that minimize collisions. Asking for the
1312 clearest path from door to window directly maps to robotic and human navigation. It is essential in
1313 autonomous driving and indoor mobility aids.

1314 **Viewpoint Visibility.** From a given viewpoint, certain objects may be blocked or visible. For
1315 example, checking what is obscured when sitting on the couch requires simulating perspective. This
1316 strengthens AR/VR realism and robotic vision.

1317 **Spatial Sequence.** Sequence reasoning identifies the order of encounters along a path. Walking
1318 from the kitchen to the balcony, one might pass the dining table first. Such reasoning mimics human
1319 navigation memory.

1320 **Accessibility Constraints.** Constraints identify which objects are blocked from access due to
1321 obstacles. For instance, a cabinet may be unreachable because of a table in front. This is crucial for
1322 assistive AI and ergonomic design.

1323

1324 **B.6 3D GEOMETRY**

1325

1326 **Volume Comparison.** Comparing volume asks which object occupies more three-dimensional space.
1327 Judging whether a box is larger than a sphere helps in packing and containment tasks. This is
1328 important for logistics and simulation.

1329 **Stability Prediction.** Stability reasoning predicts whether adding or moving objects will topple a
1330 stack. For instance, checking if placing a book will collapse a pile mirrors human intuition. This
1331 supports safety in robotics and construction.

1332 **Shape Projection.** Projection tasks ask what shape an object would cast under certain rotations
1333 or lighting. For example, a sphere always projects a circle, while a cube may project a square or
1334 hexagon. Such reasoning strengthens geometric understanding.

1335 **Spatial Containment.** Containment checks whether one object can fit inside another. For example,
1336 reasoning if a ball can fit in a box supports packing and design. This task aligns with intuitive
1337 geometry.

1338 **Gravity Effects.** Gravity reasoning predicts motion outcomes under physical forces. For example, a
1339 rolling ball may stop under a table due to friction and barriers. This strengthens real-world simulation
1340 capabilities.

1341

1342 **C MORE DETAILS ON BENCHMARK CONSTRUCTION**

1343

1344 **C.1 ANNOTATOR RECRUITMENT**

1345

1346 To support reliable quantitative comparison, we recruited annotators aged 24 to 30, all of whom
1347 were engineering graduates. Annotators participated voluntarily, worked independently, and received
1348 structured training before annotation began. The recruitment and training process emphasized
1349

consistent interpretation of spatial categories, balanced difficulty across task types, and adherence to ethical and quality-control standards. This setup helped ensure that the final annotations were reliable and aligned with the benchmark taxonomy.

C.2 ANNOTATOR TRAINING AND PROTOCOL

Before annotation, we designed a structured training program to align annotators on the six core categories (*Relative Positioning, Depth & Occlusion, Orientation, Size & Scale, Spatial Navigation, and 3D Geometry*) and their thirty subcategories. For each subcategory, annotators reviewed curated examples and counter-examples covering spatial relations such as directional relations, occlusion inference, and stability prediction. This calibration was intended to reduce category drift and make subcategory-level accuracy comparisons meaningful. Beyond the taxonomy, annotators were trained to consider six meta-dimensions of image variability: lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints. We emphasized how these factors alter empirical difficulty, for example by making occlusion harder under low contrast or scale consistency harder under perspective distortion.

We also trained annotators on the two evaluation modes used throughout the paper. In MCQ items, annotators designed distractors that test fine-grained discrimination rather than trivial elimination. In open-ended items, they wrote prompts that require compositional reasoning and visually grounded answers. Calibration sessions asked annotators to create both formats for the same images and revise them with feedback, ensuring that MCQ and open-ended scores compare the same underlying spatial skill. To situate task design within the broader research landscape, we reviewed prior benchmarks such as SpatialBot-Bench, EmbSpatial, and SpatialMM with annotators and identified their limitations. Many emphasize simplified relations, synthetic or staged scenes, or narrow reasoning types, while underrepresenting natural complexity such as shadows, reflections, and gravity-sensitive stability. This gap analysis motivated richer, more ambiguous, and more layered spatial contexts.

Finally, we provided demonstrations of the full annotation protocol: selecting images with sufficient spatial complexity, mapping each question unambiguously to a subcategory, avoiding trivial formulations, and prioritizing multi-step reasoning such as depth layering, relational chains, and cross-view scale inference. Through this training and iterative feedback, the final dataset balances taxonomic coverage, real-world relevance, and comparability across evaluation formats.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

D.1 MULTIPLE CHOICE EVALUATION

For the MCQ condition, the answer-model prompt is shown in Figure 6. We set `temperature` to 0 and `max_tokens` to 10, while keeping all other generation parameters at their default values, so the reported accuracy reflects deterministic option selection under a fixed prompt.

D.2 OPEN-ENDED EVALUATION

For the open-ended condition, Figure 6 provides both the answer-model prompt and the judge-model prompt. We set `temperature` to 0.7 and `max_tokens` to 1200, with all other parameters kept at their default values, so the metric captures free-form generation followed by a consistent automatic judgment protocol.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To validate *Gemini-2.5-Flash* as the automatic judge for open-ended scores, we selected a balanced 240-sample subset covering all categories. Each sample contained the question, ground truth, and model output, and three human annotators independently assigned binary labels (1 for correct and 0 for incorrect). We then compared LLM judgments against human annotations overall and by category using three complementary agreement metrics: **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**.

1404	
1405	SpatiaLab-MCQ : Direct : Prompt for Answering Model
1406	You are a spatial reasoning assistant. Given a multiple choice question about an image,
1407	Answer with a single number (1-4) corresponding to the option.
1408	Question:
1409	{question}
1410	Options:
1411	1. {opt1}
1412	2. {opt2}
1413	3. {opt3}
1414	4. {opt4}
1415	
1416	SpatiaLab-Open : Direct : Prompt for Answering Model
1417	You are a spatial reasoning assistant.
1418	Answer the following question in 2-4 sentences about the given image.
1419	Question:
1420	{question}
1421	
1422	SpatiaLab-Open : Direct : Prompt for Judge Model
1423	You are a strict evaluator.
1424	You will receive a question, the ground truth answer, and a model's answer.
1425	Return only 1 if the model's answer is correct, otherwise 0.
1426	Question:
1427	{question}
1428	
1429	Ground Truth Answer:
1430	{ground_truth}
1431	
1432	Model Answer:
1433	{model_answer}
1434	
1435	Output only a single digit: 0 or 1.

Figure 6: Evaluation prompts for the two scoring conditions. The prompts define the MCQ answer-selection setting and the open-ended answer/judge setting used for quantitative comparison.

Cohen’s Kappa. Cohen’s kappa (κ) measures pairwise agreement between two raters while correcting for chance agreement. For binary task, it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o is the observed agreement and p_e is the expected agreement by chance: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here, $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities that each rater assigns class c . An $\kappa = 1$: perfect agreement; $\kappa = 0$: agreement equals chance; $\kappa < 0$: systematic disagreement) We computed Cohen’s kappa between the LLM and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}), as well as between the LLM and the majority vote of the annotators (κ_{majority}).

Accuracy with Majority Vote. We also measured raw agreement between the LLM and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the LLM’s decision for sample i , and y_i^{maj} is the majority vote among annotators. Unlike Cohen’s kappa, accuracy does not correct for chance agreement; it simply reflects the proportion of identical labels.

Fleiss’ Kappa. To assess inter-annotator reliability, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to multiple raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, where n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is $\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is then computed as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Higher values indicate stronger inter-annotator reliability beyond chance.

Interpretation Table 5 shows that the LLM judge (Gemini-2.5-Flash) demonstrates strong overall agreement with human annotators, achieving a Cohen’s kappa of 0.738 with the majority vote and 0.681–0.795 against individual annotators. Raw accuracy against the majority is 0.880 overall, while Fleiss’ kappa among human annotators is 0.774, indicating substantial consistency. Certain categories, such as 3D Geometry, reach perfect agreement with $\kappa = 1.000$ and accuracy of 1.000. Depth & Occlusion also shows high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). Categories like

Table 5: Reliability of the open-ended evaluation judge. Agreement metrics compare Gemini-2.5-Flash with three human annotators across categories, reporting Cohen’s kappa against the majority vote and each annotator, raw majority-vote accuracy, and Fleiss’ kappa among annotators. Bold values mark the highest agreement within each category.

Category	κ Majority	κ Human1	κ Human2	κ Human3	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) show moderate agreement but remain within acceptable reliability ranges.

These results indicate that Gemini-2.5-Flash performs comparably to human evaluators in scoring correctness, with Cohen’s kappa values mostly in the substantial range (0.61–0.80) and some categories reaching almost perfect agreement (higher than 0.81). Fleiss’ kappa of 0.774 confirms that human annotators themselves are highly consistent, validating the reliability of the reference scores. The LLM achieves perfect or near-perfect agreement in clearly defined categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933), demonstrating accurate pattern recognition. Moderate agreement in categories like Orientation and Size & Scale reflects task complexity and inherent subjectivity rather than model shortcomings. Overall, it shows that LLM-as-judge shows robust reliability and can serve as a practical automated scoring tool, closely mirroring human evaluation.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. The closed-source group in Table 6 sets several of the strongest MCQ ceilings, but the same table also shows large within-model variation. **GPT-4o-mini** reaches 85.71% in *Stacking Orientation*, the highest subcategory score in this analysis. **Gemini-2.0-Flash** is comparatively strong on occlusion and projection, with 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*. **GPT-5-mini** adds strong orientation/depth evidence, scoring 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. These favorable peaks coexist with weaker cells, such as **Claude 3.5 Haiku** at only 32.0% in *Relative Size Comparison*. Overall, closed-source systems often outperform open-source systems by 15–25 percentage points across several categories, yet within-model spreads frequently exceed 50 percentage points, indicating higher ceilings rather than uniform spatial competence.

Open-source Large Models. Large open-source models in Table 7 improve selected subcategory ceilings while retaining sharp failure modes. **InternVL3.5-72B** reaches 80.56% in *Corner/Angle Positioning*, one of the strongest open-source MCQ results, and both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** reach 66.67% in *Complete Occlusion Inference*. **llama-3.2-90b-vision-instruct** also scores 71.43% in *Stacking Orientation*. However, the same family includes severe lows, such as **llama-3.2-11b-vision-instruct** at 2.0% in *Object Rotation*, and gravity reasoning remains modest at 56.0%. The evidence therefore favors scale as a way to raise subcategory peaks, but not as a sufficient mechanism for removing structural weaknesses; within-model spreads often exceed 70 percentage points.

Open-source Small Models. Small open-source models in Table 8 show localized strengths but the least stable coverage. **qwen-2.5-vl-7b-instruct** reaches 74.29% in *Stacking Orientation*, whereas **InternVL3.5-1B** falls to 12.0% in *Cardinal Direction*. Navigation and physics-oriented tasks remain limited: the best *Pathway Existence* score is 34.78%, and the top *Gravity Effects* score is 50.0% from

Table 6: Subcategory-level multiple-choice accuracy on SPATIALAB-MCQ by Question Sub-Categories for Closed-source Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	42.0	34.0	42.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: Subcategory-level multiple-choice accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Large Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3.5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	48.0	42.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

InternVL3.5-4B and **Gemma-3-4B-it**. These margins show that compact models can exploit strong visual templates, but their accuracy drops on tasks requiring deeper spatial abstraction. The frequent 60-point within-model spreads indicate cue-dependent competence rather than robust generalization across the taxonomy.

Reasoning Models. Reasoning-enhanced models in Table 9 provide some of the clearest evidence that explicit reasoning helps particular MCQ subcategories. **o4-mini** reaches 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*. **Gemini-2.0-Flash-Thinking** reaches 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*. The strongest reasoning-model peaks come from **Gemini-2.5-Flash-Thinking**, which scores 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. At the same time, the family remains uneven, with cells such as 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. Thus, reasoning augmentation improves selected

Table 8: Subcategory-level multiple-choice accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Small Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	34.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

Table 9: Subcategory-level multiple-choice accuracy on SPATIALAB-MCQ by Question Sub-Categories for Reasoning Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

relational and geometric comparisons, but within-model spreads above 35 percentage points show that it does not yet produce broad spatial coverage.

Spatial Reasoning Models. Spatially specialized models in Table 10 show targeted but bounded gains. **SpaceQwen2.5-VL-3B-Instruct** scores 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, indicating improved size-scale reasoning relative to many general open-source baselines. The same model leads some navigation tasks, with 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** reach 50.0% in *Stability Prediction*. However, performance drops to 24.0% in *Spatial Containment*, and these models

Table 10: Subcategory-level multiple-choice accuracy on SPATIALAB-MCQ by Question Sub-Categories for Spatial Reasoning Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5-VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

rarely exceed 55.0% even in their stronger areas. The pattern indicates that specialization raises narrow peaks but does not integrate those gains into a general spatial reasoning framework.

Overall Discussion. Across all MCQ subcategory tables (Table 8, Table 7, Table 6, Table 10, Table 9), the dominant empirical pattern is fragmented competence. Small open-source models peak at 74.29% in *Stacking Orientation* but fall to 12.0% in *Cardinal Direction*. Large models raise ceilings to 80.56% in *Corner/Angle Positioning*, but still include failures such as 2.0% in *Object Rotation*. Closed-source systems produce the highest cell, 85.71% for **GPT-4o-mini** in *Stacking Orientation*, yet still fall to 32.0% in *Relative Size Comparison*. Spatial specialists improve selected navigation and stability tasks but remain capped below 55% overall, while reasoning models show gains such as 71.79% in *Complete Occlusion Inference* alongside lows around 34.0%. The broad 85.71%–2.0% range makes the practical conclusion clear: current systems achieve impressive local peaks, but no model family delivers consistent MCQ competence across physical abstraction, orientation, and compositional spatial logic.

E.2 MODEL FAILURE ANALYSIS

Table 11 reframes the MCQ results by item difficulty, counting how many models solve each item. The distribution shows that success is highly item- and category-dependent. 3D Geometry and Size & Scale have relatively balanced distributions but still include many zero-correct items, indicating that basic geometric invariances are not reliably internalized. Depth & Occlusion has more mid-range items with 2–4 correct models, suggesting partial but inconsistent reasoning about layered scenes. Orientation remains weak on *Stacking Orientation* and *Tool Handedness*, while Relative Positioning varies between moderate success on *Proximity Gradients* and many near-zero outcomes for *Corner/Angle Positioning*. Spatial Navigation shows comparatively stronger cells in *Obstacle Avoidance* and *Viewpoint Visibility*, but *Pathway Existence* remains difficult when relational chaining is required. Aggregated across bins, no category exhibits uniformly high accuracy, confirming that MCQ success is localized rather than general.

Table 11: Item difficulty distribution for SPATIALAB-MCQ, grouped by how many evaluated models (0–5) answered each item correctly across categories and subcategories. “0” marks items missed by all models, while “5” marks items solved by all five models in this analysis.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

These MCQ failures point to representational and reasoning bottlenecks rather than isolated annotation issues. Low scores in *Stacking Orientation* and *Tool Handedness* indicate limited grounding in embodied and physics-based reasoning. Errors in *Corner/Angle Positioning* and *Pathway Existence* indicate weak multi-step relational chaining. Failures on *Transparency Effects* and *Reflective Surfaces* show that non-local visual cues remain difficult. Conversely, stronger *Obstacle Avoidance* performance may reflect shortcut learning from common walkable-path patterns rather than genuine planning. The quantitative distribution therefore supports the same conclusion as the accuracy tables: models benefit from scale, proprietary data, reasoning modules, and specialization in different places, but the remaining margins require training and architectures that explicitly model reference frames, occlusion layers, and physical constraints.

Table 12: Subcategory-level open-ended accuracy on SPATIALAB-OPEN by Question Sub-Categories for Closed-source Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	<u>48.0</u>	36.0	30.0	24.0	20.0
	Shape Projection	<u>40.0</u>	18.0	22.0	16.0	16.0
	Spatial Containment	<u>50.0</u>	38.0	<u>50.0</u>	36.0	36.0
	Stability Prediction	<u>50.0</u>	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	<u>41.3</u>	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	<u>46.15</u>	35.9	38.46	28.21	25.64
	Layering Order	<u>30.0</u>	22.86	21.43	17.14	15.71
	Partial Occlusion	<u>38.0</u>	20.0	28.0	14.0	14.0
	Reflective Surfaces	<u>32.0</u>	28.0	<u>32.0</u>	22.0	22.0
	Transparency Effects	<u>32.0</u>	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	<u>40.0</u>	34.0	26.0	30.0
	Facing Direction	<u>28.57</u>	<u>28.57</u>	26.19	19.05	14.29
	Object Rotation	<u>42.0</u>	14.0	28.0	30.0	14.0
	Stacking Orientation	<u>37.14</u>	20.0	31.43	22.86	28.57
	Tool Handedness	<u>48.0</u>	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	<u>55.81</u>	34.88	37.21	37.21	34.88
	Betweenness Relationships	<u>48.94</u>	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	<u>44.44</u>	30.56	41.67	16.67	36.11
	Directional Relations	<u>58.14</u>	46.51	<u>58.14</u>	34.88	37.21
	Proximity Gradients	<u>39.53</u>	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	<u>48.0</u>	32.0	36.0	20.0	22.0
	Perspective Distortion	<u>48.0</u>	24.0	22.0	20.0	12.0
	Relative Size Comparison	<u>36.0</u>	26.0	28.0	18.0	16.0
	Scale Consistency	<u>43.14</u>	21.57	23.53	19.61	9.8
	Shadow-Size Projection	<u>37.25</u>	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	<u>41.46</u>	31.71	39.02	24.39	21.95
	Obstacle Avoidance	<u>40.0</u>	22.0	22.0	26.0	14.0
	Pathway Existence	<u>23.91</u>	15.22	21.74	10.87	19.57
	Spatial Sequence	<u>52.0</u>	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	<u>32.0</u>	20.0	10.0

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). Closed-source models lead the open-ended setting overall, but their advantage is uneven across subcategories. GPT-5-mini is the strongest model in this group, with peaks of 58.14% in Directional Relations and 55.81% in Alignment Patterns, giving it the clearest evidence of relative strength in **3D Geometry** and **Relative Positioning**. The margins narrow sharply in harder continuous-reasoning tasks: Gemini-2.5-Flash records 16.28% in Proximity Gradients, and Claude 3.5 Haiku records 12.0% in Perspective Distortion. This pattern shows that proprietary training and scale improve open-ended ceilings, but do not eliminate failures on graded spatial cues. Closed-source dominance is therefore relative rather than absolute.

Open-source Large Models (Table 13). Large open-source models show a distributed performance profile, with different systems leading isolated subcategories. llama-3.2-90b-vision-instruct reaches 60.0% in Depth and Occlusion, and GLM-4.5V-106B-MoE leads Spatial Containment at 38.0%. These favorable peaks coexist with large failures: qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct both score only 9.3% in Proximity Gradients. The comparison suggests that scale can improve specific depth or containment judgments, but it does not guarantee coverage of subtle spatial analogies or distance scaling. Inter-model variance is also high, indicating that training-data composition and architecture affect open-ended robustness.

Open-source Small Models (Table 14). Small open-source models generally lag larger systems, but their subcategory results still contain useful localized evidence. qwen-2.5-vl-3b-instruct reaches 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads Spatial Containment at 24.0% and Stability Prediction at 21.43%. These pockets of strength are offset by severe failures: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B reaches only 2.0% in Shape Projection. The resulting margins indicate that compact models can learn narrow spatial cues but lack the capacity and training diversity needed for consistent open-ended reasoning.

Table 13: Subcategory-level open-ended accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Large Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	28.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

Table 14: Subcategory-level open-ended accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Small Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	17.39	8.7	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

Reasoning Models (Table 15). Reasoning models show the largest internal contrast. Gemini-2.5-flash-thinking is the strongest member of the group, with 75.0% in Tool Handedness and 57.14% in Complete Occlusion Inference, demonstrating that explicit reasoning can substantially improve selected open-ended spatial judgments. Kimi-VL-A3B-Thinking-2506, however, scores only 6.0% in Spatial Containment and lags across many subcategories. The family-level evidence therefore supports reasoning as a useful intervention, but only when it is integrated with sufficiently strong multimodal grounding. Even Gemini-2.5-flash-thinking does not dominate Shape Projection or Betweenness Relationships, showing that reasoning modules remain incomplete rather than universally corrective.

Spatial Reasoning Models (Table 16). Spatial reasoning models perform weakest in the open-ended condition despite being explicitly designed for spatial tasks. Several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction. SpaceQwen2.5-VL-3B-Instruct is slightly stronger in selected cells, including 23.91% in Volume Comparison and 13.73% in Scale Consistency, but these scores remain far below the strongest closed-source and reasoning-model results. Betweenness Relationships and Corner/Angle Positioning collapse to 0.0%

Table 15: Subcategory-level open-ended accuracy on SPATIALAB-OPEN by Question Sub-Categories for Reasoning Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	<u>40.0</u>	14.0	38.0
	Shape Projection	24.0	0.0	10.0	<u>30.0</u>
	Spatial Containment	36.0	<u>50.0</u>	6.0	30.0
	Stability Prediction	26.19	<u>42.86</u>	14.29	33.33
	Volume Comparison	36.96	<u>44.44</u>	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	<u>57.14</u>	17.95	33.33
	Layering Order	<u>20.0</u>	<u>20.0</u>	14.29	14.29
	Partial Occlusion	28.0	<u>50.0</u>	8.0	14.0
	Reflective Surfaces	34.0	<u>63.64</u>	10.0	16.0
	Transparency Effects	26.0	<u>57.14</u>	22.0	20.0
Orientation	Cardinal Direction	<u>40.0</u>	20.0	18.0	26.0
	Facing Direction	35.71	<u>37.5</u>	14.29	28.57
	Object Rotation	18.0	<u>40.0</u>	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	<u>34.29</u>
	Tool Handedness	44.0	<u>75.0</u>	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	<u>66.67</u>	25.58	41.86
	Betweenness Relationships	<u>27.66</u>	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	<u>50.0</u>	11.11	11.11
	Directional Relations	39.53	<u>44.44</u>	27.91	37.21
	Proximity Gradients	<u>23.26</u>	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	<u>57.14</u>	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	<u>30.0</u>
	Relative Size Comparison	28.0	11.11	14.0	<u>36.0</u>
	Scale Consistency	<u>33.33</u>	20.0	11.76	31.37
	Shadow-Size Projection	<u>35.29</u>	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	<u>45.45</u>	17.07	41.46
	Obstacle Avoidance	<u>26.0</u>	10.0	16.0	18.0
	Pathway Existence	23.91	<u>25.0</u>	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	<u>36.0</u>
	Viewpoint Visibility	<u>32.0</u>	12.5	16.0	26.0

and 2.78%, respectively. These numbers indicate that narrow spatial specialization does not transfer reliably to free-form generation without sufficient capacity, data diversity, and linguistic grounding.

Overall Discussion. Across closed-source (Table 12), open-source large (Table 13), open-source small (Table 14), reasoning (Table 15), and spatial reasoning (Table 16) groups, open-ended evaluation reveals even stronger fragmentation than MCQ. GPT-5-mini reaches 58.14% in Directional Relations and 55.81% in Alignment Patterns, but closed-source systems still struggle on Proximity Gradients. Large open-source models can reach 60.0% in Depth and Occlusion, yet fall to 9.3% in Proximity Gradients. Small models show rare strengths such as 26.0% in Relative Size Comparison but collapse to 0.0% in Betweenness Relationships. Reasoning helps most clearly in Gemini-2.5-flash-thinking’s 75.0% Tool Handedness result, but other reasoning models remain much weaker. Spatial specialists often remain below 15% and fail completely on core relational tasks. The quantitative evidence points to three recurring causes: missing representations for continuous spatial cues, limits tied to model scale and training diversity, and brittle reasoning pipelines that do not generalize across subcategories.

F.2 MODEL FAILURE ANALYSIS

Table 17 shows that open-ended item difficulty is skewed toward the lower bins, where 0–2 models answer correctly. 3D Geometry is broadly challenging, with Shape Projection producing 19 zero-correct items and Stability Prediction producing 10. Depth and Occlusion is the hardest category by zero-correct count, accumulating 72 such items, including 20 in Layering Order and 16 in Transparency Effects. Orientation also shows persistent weaknesses: Stacking Orientation has 11 zero-correct items, and Tool Handedness has 5 zero-correct items with no perfect-success items. Relative Positioning is inconsistent, with Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models. Size and Scale is comparatively stronger in places, including Scale Consistency with 6 full-success items, but Shadow-Size Projection still has 17 zero-correct items. Spatial Navigation is split between occasional full-model success in Spatial Sequence and Obstacle Avoidance and high failure counts in Pathway Existence (16 zero) and Accessibility Constraints (10 zero). Overall, zero-correct items dominate the distribution, while only 74 items are solved by all five models.

Table 16: Subcategory-level open-ended accuracy on SPATIALAB-OPEN by Question Sub-Categories for Spatial Reasoning Models. Bold and underlined values mark the best subcategory score within the corresponding model group.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	<u>16.0</u>
	Shape Projection	<u>8.0</u>	<u>8.0</u>	6.0
	Spatial Containment	8.0	<u>10.0</u>	<u>10.0</u>
	Stability Prediction	<u>14.29</u>	<u>14.29</u>	7.14
	Volume Comparison	19.57	<u>23.91</u>	<u>23.91</u>
Depth and Occlusion	Complete Occlusion Inference	7.69	<u>10.26</u>	7.69
	Layering Order	8.57	<u>11.43</u>	4.29
	Partial Occlusion	10.0	<u>12.0</u>	4.0
	Reflective Surfaces	<u>4.0</u>	<u>4.0</u>	2.0
	Transparency Effects	4.0	<u>8.0</u>	2.0
Orientation	Cardinal Direction	18.0	<u>24.0</u>	8.0
	Facing Direction	<u>23.81</u>	<u>23.81</u>	16.67
	Object Rotation	<u>20.0</u>	<u>20.0</u>	18.0
	Stacking Orientation	8.57	8.57	<u>17.14</u>
	Tool Handedness	0.0	4.0	<u>8.0</u>
Relative Positioning	Alignment Patterns	<u>32.56</u>	<u>32.56</u>	23.26
	Betweenness Relationships	<u>4.26</u>	2.13	0.0
	Corner/Angle Positioning	<u>2.78</u>	<u>2.78</u>	<u>2.78</u>
	Directional Relations	<u>11.63</u>	6.98	<u>11.63</u>
	Proximity Gradients	6.98	6.98	<u>9.3</u>
Size and Scale	Distance-Size Correlation	<u>18.0</u>	<u>18.0</u>	16.0
	Perspective Distortion	16.0	<u>20.0</u>	2.0
	Relative Size Comparison	<u>22.0</u>	20.0	8.0
	Scale Consistency	9.8	11.76	<u>13.73</u>
	Shadow-Size Projection	<u>27.45</u>	<u>27.45</u>	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	<u>9.76</u>
	Obstacle Avoidance	10.0	8.0	<u>14.0</u>
	Pathway Existence	6.52	6.52	<u>8.7</u>
	Spatial Sequence	<u>16.0</u>	12.0	<u>16.0</u>
	Viewpoint Visibility	<u>20.0</u>	16.0	8.0

The open-ended failure distribution indicates that the main bottleneck is representational and generative rather than simply dataset design. Depth and Occlusion failures suggest reliance on visible surface features instead of structured scene representations. Tool Handedness failures expose missing embodied affordance modeling. Relative Positioning and Spatial Navigation errors indicate weak multi-hop relational chaining, especially when Pathway Existence requires local connectivity to be integrated with global layout. Partial success in Size and Scale suggests that models sometimes exploit perspective or scale regularities, but failures on shadows and less common configurations show that these cues are not robust. The resulting pattern is consistent with the main paper: current VLMs succeed when salient cues align with training priors, but open-ended spatial reasoning breaks down when answers require counterfactual, compositional, or embodied inference.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

This appendix quantifies how much performance changes when the same spatial reasoning benchmark is evaluated with multiple-choice (MCQ) answers versus open-ended generation. Table 18 reports the category-level gaps, making the comparison explicit by model, spatial category, and answer format.

G.1 QUANTITATIVE SNAPSHOT

Across 25 models, the average MCQ–open-ended gap is **23.0%** with a standard deviation of 5.5%. The subtask means are tightly clustered: 3D geometry (24.57), depth & occlusion (23.45), orientation (23.70), relative positioning (23.11), size & scale (23.42), and spatial navigation (22.89), for an overall average of 23.01. At the model level, the overall gap ranges from 11.93 to 31.57, with a median of approximately 23.34. Spatial reasoning models have the largest family-level gap (27.03), whereas reasoning-oriented models have the smallest (19.11). Proprietary models average lower gaps (21.09) than many long open-source models (24.51). The largest individual gaps are InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%); the smallest are Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36).

Table 17: Item difficulty distribution for SPATIALAB-OPEN, grouped by how many evaluated models (0–5) answered each item correctly across categories and subcategories. “0” marks items missed by all models, while “5” marks items solved by all five models in this analysis.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

Correlations show that spatial navigation is the strongest driver of the overall gap (Pearson $r = 0.99$), followed by orientation ($r = 0.83$) and 3D geometry ($r = 0.79$).

G.2 PATTERNS AND OBSERVATIONS

Three patterns structure the evidence. First, specialist spatial reasoning models show the largest MCQ→open-ended gaps, suggesting that their strengths are more aligned with classification-style supervision than with free-form generation. Reasoning-focused models show smaller gaps, indicating that chain-of-thought-style training can stabilize generative answers. Second, spatial navigation is the dominant contributor to the aggregate disparity: models that lose accuracy on open-ended navigation also tend to have the largest overall gaps. Orientation and 3D geometry contribute strongly,

Table 18: **MCQ→open-ended performance gap (%) (↓) across spatial categories on SPATIALAB.** A lower value indicates more consistent performance across answer formats, whereas a higher value identifies stronger format sensitivity. We bold and underline the highest gap within each model group.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

while relative position and size/scale vary more by model. Third, scale does not by itself predict robustness: Llama-3.2-11B has one of the smallest gaps, whereas InternVL3.5-72B has the largest. Negative or near-zero gaps, including depth & occlusion for Llama-3.2-11B at -1.98 and relative position for o4-mini at -3.18 , show that MCQ distractors can sometimes depress or distort measured performance.

G.3 HYPOTHESES ON ROOT CAUSES

We hypothesize several interacting factors behind the observed MCQ→open-ended gaps, each tied to the statistics above:

1. **MCQ structural advantage.** MCQ constrains the output space, allowing models to exploit surface cues or eliminate distractors. Across models, MCQ scores are systematically higher than open-ended scores, with an average gap of 23.0% and $\sigma = 5.5\%$. Negative gaps for Llama-3.2-11B in depth & occlusion (-1.98) and o4-mini in relative position (-3.18) show that open-ended generation can occasionally be more faithful when distractors are misleading.

Recommendation: Report MCQ scores together with open-ended scores and subtask diagnostics to reduce overestimation from option-based cues.

2. **Specialization bias in spatial reasoning models.** Spatial reasoning models have the largest average gap (27.03%) despite being optimized for spatial tasks, suggesting reliance on categorical or synthetic selection supervision. InternVL3.5-72B shows the highest overall gap (31.57%), especially in spatial navigation (36.68%) and orientation (34.44%). Correlations between subtask gaps and the overall gap (SpNav: $r = 0.99$, Orientation: $r = 0.83$) indicate that these format-dependent failures are concentrated in sequential and geometric reasoning tasks.

Recommendation: Add open-ended spatial generation to fine-tuning and instruction-tuning so that spatial specialists are less dependent on categorical supervision.

-
3. **Instruction-tuning and stepwise decoding.** Reasoning-oriented models, including Gemini-2-Flash-Thinking and o4-mini, show smaller average gaps (around 19.11%) and lower subtask variance. Gemini-2-Flash-Thinking has an overall gap of 13.0%, with moderate subtask gaps such as 13.9% in 3D Geometry and 21.03% in Relative Positioning. This pattern suggests that stepwise decoding improves stability in open-ended responses.
Recommendation: Use chain-of-thought, stepwise decoding, and instruction tuning for open-ended spatial tasks when the goal is to reduce format sensitivity.
4. **Spatio-navigation stresses sequential grounding.** Spatial navigation gaps are the strongest predictor of overall MCQ→open-ended disparity (Pearson $r = 0.99$). SpaceQwen2.5-VL-3B-Instruct has a Spatial Navigation gap of 35.87% and an overall gap of 29.78%, showing that multi-step referential reasoning is disproportionately affected by the open-ended format.
Recommendation: Include dedicated open-ended spatio-navigation tasks and targeted multi-step supervision to measure and improve sequential grounding.
5. **MCQ distractor artifacts.** Some negative or near-zero gaps show that distractors can suppress measured MCQ performance, as in o4-mini’s relative-position gap (−3.18%). Conversely, very high gaps such as Qwen-VL2.5-32B-Instruct overall = 29.85% show that MCQ options can also exaggerate selection-based competence.
Recommendation: Audit MCQ distractors and pair them with open-ended evaluation to obtain less biased estimates of spatial reasoning.

G.4 PRACTICAL IMPLICATIONS

The practical message is that MCQ and open-ended scores should be reported as a pair. MCQ-only reporting can inflate estimates of usable spatial competence by about 23% on average, and the size of the inflation depends strongly on subtask, especially spatial navigation. Per-subtask breakdowns, prompt-controlled open-ended evaluation, and distractor audits provide a clearer estimate of which models are genuinely improving and which are benefiting from format-specific cues.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To quantify which interventions improve spatial reasoning, we evaluate several complementary strategies under the same category and format structure used in the main benchmark.

H.1 INHERENT REASONING MECHANISMS

We first compare standard models with their reasoning-enabled counterparts (Table 2, Table 3). In MCQ evaluation, reasoning improves accuracy in several categories, with the clearest overall margin for Gemini-2.5-Flash-Thinking over Gemini-2.5-Flash (+4.64%; 52.93% vs. 48.29%). The category gains are especially visible in Relative Positioning (+13.1%) and Orientation (+8.6%). o4-mini-medium reaches 53.21% overall MCQ accuracy, further showing that explicit reasoning scaffolds are most favorable in structured, discrete-answer settings.

Open-ended evaluation preserves the same direction but with smaller and less uniform margins. Gemini-2.5-Flash-Thinking improves slightly over Gemini-2.5-Flash (32.77% vs. 30.93%), while o4-mini-medium reaches 37.86%, surpassing the non-reasoning models in this setting. These results indicate that reasoning mechanisms can stabilize free-form generation, particularly for complex categories such as 3D Geometry and Spatial Navigation. However, the gains are not universal: Gemini-2.5-Flash-Thinking drops in Size & Scale (21.74% vs. 26.59%), reinforcing that generative spatial reasoning remains brittle even when overall reasoning improves.

Overall, the comparison is favorable to reasoning modules for structured MCQ accuracy and partially favorable for open-ended stability, but the effect is category-dependent. Reasoning strengthens logical consistency and relational chaining, yet the remaining margins in scale, occlusion, and free-form grounding show that reasoning is not a substitute for perceptual spatial representation.

2106
2107
2108
2109
2110
2111
2112
2113
2114
2115
2116
2117
2118
2119
2120
2121
2122
2123
2124
2125
2126
2127
2128
2129
2130
2131
2132
2133
2134
2135
2136
2137
2138
2139
2140
2141
2142
2143
2144
2145
2146
2147
2148
2149
2150
2151
2152
2153
2154
2155
2156
2157
2158
2159

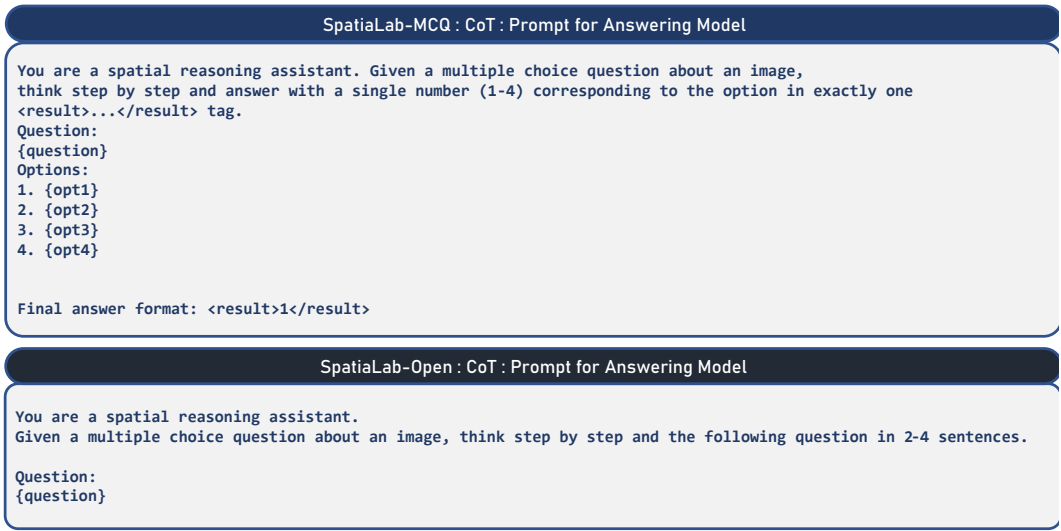


Figure 7: Chain-of-thought (CoT) prompts used to test whether step-by-step reasoning changes category-level accuracy.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To test whether structured reasoning changes spatial accuracy, we evaluate Chain-of-Thought (CoT) prompting on a balanced subset. For each of the 30 subcategories in SPATIALAB, we sample 5 items with seed 42, yielding coverage of all major spatial phenomena while keeping the evaluation tractable. Figure 7 shows the prompts. Because the model and sampled items are fixed, the reported gains isolate the effect of CoT prompting from model scale or training differences.

Table 19: Category-level effect of CoT prompting for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%), and Gains report the with-CoT minus without-CoT margin for each category.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

The results in Table 19 show a narrow but interpretable benefit: Orientation is the only category where CoT reliably improves accuracy, while most other categories stagnate or decline. This pattern suggests that CoT helps when the task reduces to directional alignment, but it does not supply the missing perceptual priors needed for depth, scale, or relative-position judgments. When those priors are wrong, step-by-step reasoning tends to propagate the error rather than correct it.

This finding highlights the practical significance of the benchmark: aggregate spatial performance cannot be improved reliably by simply adding multi-step text reasoning. Unlike many textual reasoning tasks, the categories in SPATIALAB expose whether the model has grounded perceptual evidence. In depth and size tasks, incorrect visual priors make additional reasoning steps harmful, and the sharp drops for Gemini in MCQ settings show how fragile reasoning becomes without robust perceptual support.

Thus, the CoT comparison strengthens the benchmark evidence: it separates improvements from verbal reasoning scaffolds from improvements that require grounded geospatial and 3D representations. Solving SPATIALAB requires more than chaining logic; it requires embeddings and training signals that preserve the perceptual structure needed for spatial cognition.

2160
2161
2162
2163
2164
2165
2166
2167
2168
2169
2170
2171
2172
2173
2174
2175
2176
2177
2178
2179
2180
2181
2182
2183
2184
2185
2186
2187
2188
2189
2190
2191
2192
2193
2194
2195
2196
2197
2198
2199
2200
2201
2202
2203
2204
2205
2206
2207
2208
2209
2210
2211
2212
2213

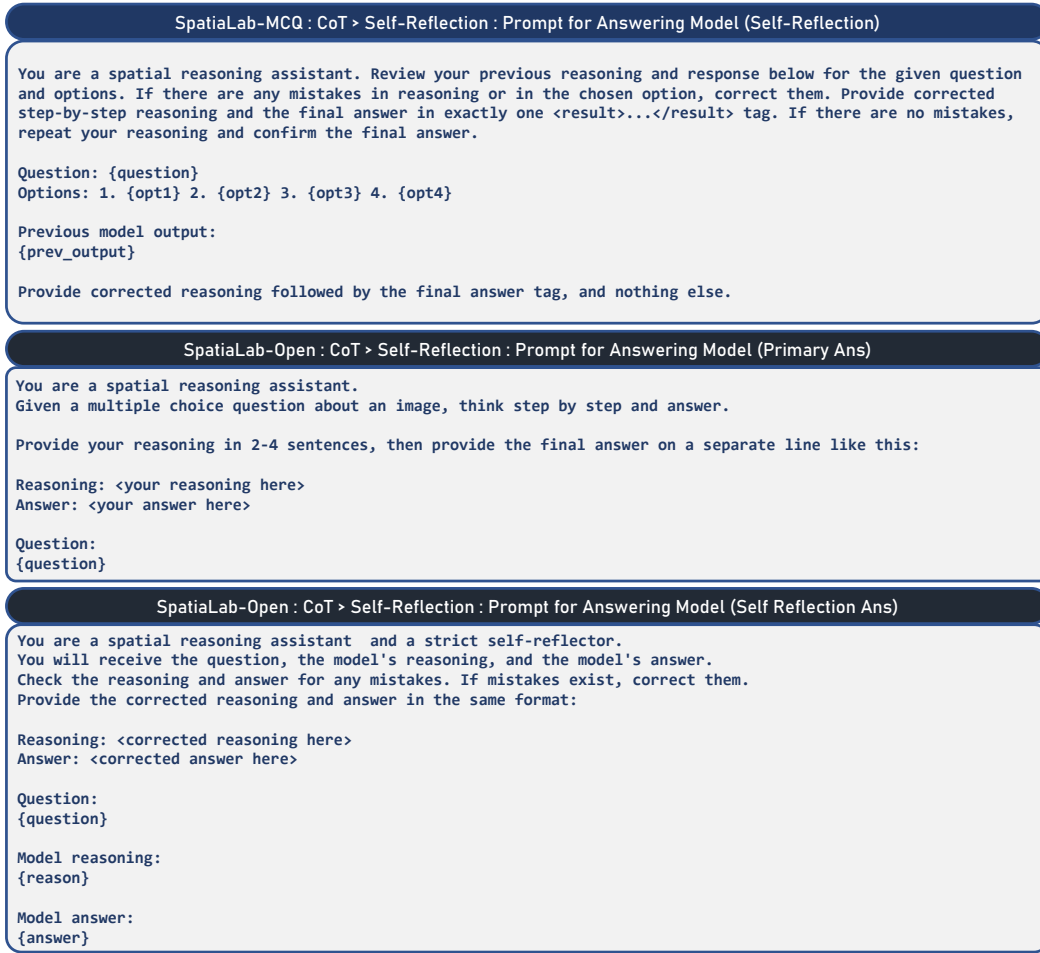


Figure 8: Chain-of-thought (CoT) with Self-Reflection Prompts.

H.3 CHAIN-OF-THOUGHT (CoT) PROMPTING WITH SELF-REFLECTION

We further extend the CoT setup by adding a self-reflection stage, where the model is prompted to review and revise its initial reasoning before finalizing an answer (see Figure 8). The evaluation protocol remains identical to the standard CoT setting, using 5 balanced samples per sub-category (seed 42). This design isolates the effect of self-reflection while keeping the comparison directly aligned with baseline CoT performance.

Table 20: Category-level comparison of CoT and CoT + Self-Reflection (CoT-SR) for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%), and Gains report the CoT-SR minus CoT margin.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

Adding self-reflection to Chain-of-Thought (CoT) prompting produced mixed and highly model-dependent outcomes. For InternVL-3-78b, overall accuracy declined slightly (-2.66% MCQ, -8.67%

open-ended), with sharp drops in categories like depth, relative positioning, and navigation, suggesting that reflection often reinforced errors instead of correcting them. Some localized gains were observed, such as +12% in Size & Scale (MCQ), but these were inconsistent and offset by large degradations (−28% in relative positioning open-ended). In contrast, Gemini-2.5-Flash showed strong MCQ gains (+20.66%), with particularly large improvements in 3D Geometry (+32%) and Depth & Occlusion (+24%), though open-ended improvements remained negligible (+0.67%). These results highlight a key asymmetry: reflection helps in multiple-choice contexts where models can prune unlikely options, but does not transfer to free-form generation where uncertainty propagates into ungrounded rationales. We also find that the categories most improved by reflection are those where geometric consistency is central (orientation, depth, size), while tasks requiring relational integration across multiple objects (relative positioning, navigation) degrade. Overall, reflection functions as a stabilizer in structured choice settings but fails to reliably enhance naturalistic reasoning.

This observed inconsistency mirrors our earlier findings with plain CoT prompting: reasoning enhancements only succeed when the model has stable perceptual anchors to ground its steps. Self-reflection adds an explicit verification pass, but this mechanism is only as strong as the model’s internal priors. When depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass amplifies misconceptions rather than repairing them. This explains why Gemini, which exhibits stronger perceptual grounding, benefits under MCQ conditions, while InternVL-3-78b suffers more frequent degradations. The broader root cause lies in a pretraining mismatch: current VLMs import reasoning templates from language modeling but lack the embedded 3D and geospatial representations that humans leverage. Without these anchors, reflection cannot perform true error correction; instead, it functions as a linguistic filter that polishes reasoning syntax without improving perceptual grounding. As a result, reflective reasoning loops provide modest utility in discrete answer formats but collapse under open-ended demands where factual verification is absent.

These findings confirm that our benchmark shows a fundamental gap in current architectures: surface-level reasoning layers alone cannot compensate for missing spatial cognition, and meaningful progress will require deeper integration of geometric and perceptual structure. Moreover, because VLLMs are commonly trained on web data, the concepts that are well-represented online are those where models tend to perform better, whereas poorly represented spatial and 3D concepts suffer, reinforcing these performance disparities.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) using a parameter-efficient approach with LoRA (Hu et al., 2021). To accommodate limited GPU resources, we loaded the model in 4-bit precision (`load_in_4bit=True`) and inserted LoRA adapters into attention and feed-forward network projection layers. The LoRA hyperparameters, including rank, scaling factor, dropout, and target modules, are summarized in Table 21. Gradient checkpointing was enabled to reduce memory usage, and a fixed random seed ensured reproducibility.

For tokenization, we followed the Qwen-2.5 chat template to maintain consistent formatting between training and inference. Each training example was converted into a single text sequence, and dynamic padding was applied with `DataCollatorForSeq2Seq`. Training used an effective batch size of 4 (per-device batch size 1 with gradient accumulation of 4). Trainer and optimization hyperparameters, including learning rate, optimizer, weight decay, and scheduler, are listed in Table 22. The training loop consisted of four passes, each spanning a single epoch, yielding four saved checkpoints.

Overall, this setup allowed us to efficiently fine-tune a large vision-language model on limited GPU resources while maintaining high adaptation capacity.

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trends in Figure 4a reveal notable differences between the MCQ and Open-ended evaluation modes over training epochs. We observe that MCQ accuracy begins at a lower baseline of 25.63% but rises consistently, converging near 36.59% by the fourth epoch. This steady improvement suggests that the model benefits strongly from additional supervised fine-tuning, particularly in tasks

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

that constrain answers within a limited candidate space. By contrast, Open-ended evaluation starts higher at 34.40% but declines sharply by the second epoch, dropping to as low as 12.62%. Only in later epochs does it partially recover, reaching 35.48% at the end. This instability points to a sensitivity in open-ended generation that is not present in the MCQ setting, where the structure of multiple-choice questions provides more stable optimization signals. We interpret the fluctuations in open-ended performance as evidence that fine-tuning may overfit to MCQ-like reasoning patterns at the expense of free-form language generalization. The dotted baselines in the plot further highlight how the MCQ mode achieves gains beyond its initial standing, whereas Open-ended remains approximately stagnant. Overall, the figure illustrates that SFT predominantly favors MCQ tasks in terms of stability and consistent accuracy gains.

The tabulated results in Table 4b complement the figure by providing a detailed before-and-after breakdown across six spatial reasoning categories. Across MCQ tasks, we find that performance improved in every category, with gains ranging from a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale. The Grand Total gain for MCQ stands at 10.97%, confirming that the fine-tuning procedure effectively enhances structured-answer performance. In contrast, Open-ended results are far more mixed, with both positive and negative changes. Categories such as 3D Geometry and Spatial Navigation show small but positive gains (3.50% and 2.82%, respectively), yet Size and Scale exhibits a significant drop of -3.95% . Relative Positioning also decreases by -0.79% , reflecting that fine-tuning can inadvertently harm certain reasoning modes in free-form contexts. The overall Open-ended gain is only 1.07%, which is statistically negligible compared to MCQ improvements. These results confirm that the effectiveness of SFT is highly sensitive to the evaluation paradigm. By isolating the categories, we underscore where training helps, where it fails, and how performance benefits differ between constrained and unconstrained outputs.

A deeper inspection reveals striking disparities between categories and formats. For MCQ, Size and Scale and Spatial Navigation yield the highest improvements, suggesting that models trained with multiple-choice prompts are especially effective at learning tasks requiring scale discrimination and navigational reasoning. However, in Open-ended form, Size and Scale actually declines, showing that the same learned representations do not transfer well to generative contexts. Similarly, Relative Positioning barely improves in MCQ and worsens in Open-ended, highlighting a consistent weakness

in relational reasoning regardless of format. Conversely, Orientation shows strong MCQ gains (+13.33%) and small Open-ended improvements, suggesting that some categories generalize more robustly. We hypothesize that the divergence between the two formats may stem from the structural scaffolding MCQs provide, which reduces the need for the model to generate precise linguistic formulations. Open-ended tasks, by contrast, require simultaneous mastery of reasoning and natural language fluency, which amplifies weaknesses. The trends indicate that while fine-tuning strongly enhances pattern recognition in discrete-choice settings, it introduces volatility and even degradation when free expression is required. This duality makes clear that training interventions optimized for MCQs may not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we show that SFT as currently applied provides strong benefits for structured tasks but limited or even negative transfer to open-ended reasoning. This limitation is not only an artifact of fine-tuning, but also symptomatic of how contemporary VLMs are pretrained. Existing pretraining pipelines heavily emphasize broad image-text alignment, object recognition, and captioning, yet they provide little explicit supervision on compositional spatial structures, occlusion layers, or geometric transformations. As a result, models learn to excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning suggests that the model’s learned priors are fragile: without multiple-choice scaffolding, they revert to shallow pattern-matching instead of grounded inference. Categories such as Size and Scale and Relative Positioning, where declines are most pronounced, point to missing inductive biases in pretraining, biases that humans develop through embodied interaction with 3D environments. Another contributing factor is the reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, preventing models from generalizing to real-world clutter, noise, and complexity. These findings underscore the need to rethink both pretraining and fine-tuning for spatial reasoning, incorporating datasets and objectives that reflect naturalistic visual-spatial challenges. Ultimately, we argue that without targeted reforms, current VLMs risk reinforcing an algorithmic straightjacket: they appear competent under constrained evaluations yet remain brittle and unaligned with the embodied, flexible reasoning that humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To investigate the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, reveals a distinct divergence in performance dynamics between Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As illustrated in Table 23 and Figure 9, performance on SPATIALAB-MCQ exhibits a monotonic improvement, rising from a baseline of 25.63% to an average of 35.74% at Epoch 4. This indicates that the model successfully adapts to the discriminative nature of the task, learning to map visual features to constrained output options with increasing reliability.

In sharp contrast, SPATIALAB-OPEN displays a pronounced *U-shaped* trajectory across all seeds. Rather than steady improvement, the model suffers an immediate and significant performance collapse in the early epochs (dropping from 34.40% to an average of 19.26% at Epoch 1). While performance recovers in later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible (+0.42%).

Representational Brittleness and Alignment Tax. We attribute this phenomenon to the fragility of spatial representations in current VLMs when subjected to fine-tuning. The initial collapse in open-ended accuracy suggests *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation. As the model optimizes for the specific instruction-following format of the training data, it overfits to the structural logic of the tasks (benefiting MCQ) at the expense of the flexible, generative spatial grounding required for open-ended descriptions.

The subsequent recovery phase represents the model re-learning to articulate spatial concepts within the new distribution, yet it fails to surpass the initial baseline significantly. This dynamic underscores a critical finding: SFT is effective for teaching models to *select* correct spatial answers, but it does not

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). While MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

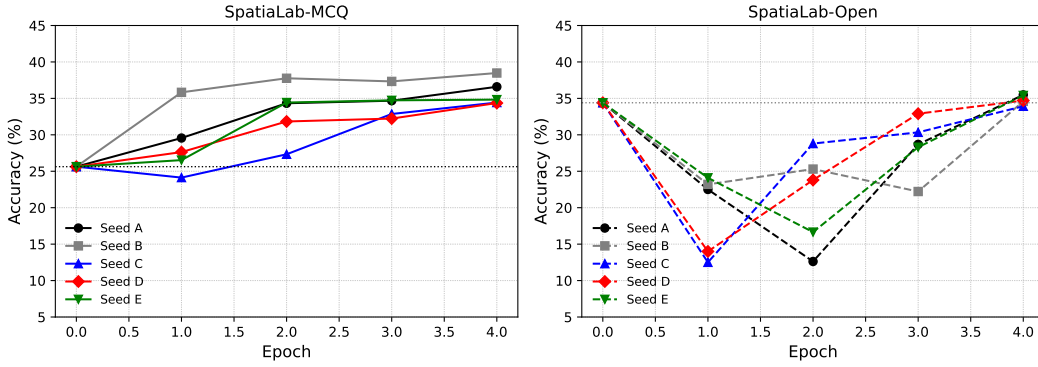


Figure 9: SFT Dynamics Analysis.

fundamentally enhance the underlying *generative spatial representation* needed for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To assess whether the representations learned from SPATIALAB generalize beyond the benchmark’s specific format, we evaluated the transferability of our fine-tuned model to three external spatial reasoning datasets: OMNISpatial, SPACE (Multimodal), and MIND THE GAP. We employed the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALAB via the SFT protocol detailed in Appendix H.4.1, and evaluated it on these external benchmarks in a zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model learns generalizable spatial representations rather than merely overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As shown in Table 24, fine-tuning on SPATIALAB yields substantial and consistent performance improvements across all evaluated domains, ranging from +5.24% to +11.70%. The most significant

gain was observed on MIND THE GAP (+11.70%), suggesting that our benchmark’s emphasis on occlusion and spatial continuity effectively addresses the reasoning gaps targeted by that dataset. Similarly, improvements on OMNISPIATIAL (+7.05%) and SPACE (+5.24%) confirm that the geometric and relational concepts encoded in our taxonomy align with broader definitions of spatial intelligence in the literature. These results validate the quality of the SPATIALAB data. While the primary purpose of this work is to serve as a rigorous diagnostic benchmark, the strong transferability demonstrates that the dataset does not merely encourage overfitting to a specific question template. Instead, the diversity of the 30 sub-categories forces the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images within the SPATIALAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each managed by a dedicated agent, allowing for focused processing of objects, attributes, relations, symmetries, and transformations. This modular design not only improves accuracy and interpretability but also supports multi-step reasoning, enabling the system to propagate low-level perceptual cues through higher-order relational and transformational inferences. The architecture, agent implementation, and their coordinated interaction are detailed in the following sections, with the full set of prompts provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including geometric shapes, natural elements, and artificial icons—alongside attributes such as size, orientation, color, and shading. Relative positions like leftmost, center, and top are encoded explicitly, and the prompt emphasizes exhaustive detail without summarization. Images are converted to data URIs and sent with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list using regex-based extraction. This structured description serves as the foundation for downstream agents, providing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features, such as partial occlusions or minor shape variations, are retained for later relational and transformational reasoning.

Object Segmentation Agent. Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, differentiating between visually similar instances. Each object receives a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent uses carefully formatted prompts and calls the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation reduces ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent also facilitates detection of overlaps, repetitions, and hierarchical groupings that may not be evident in raw descriptions.

Attribute Extraction Agent. Attribute Extraction agent translates each object into a structured JSON capturing core properties such as shape, size, color, shading, orientation, and positional descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults to ensure robustness against model inconsistencies. This structured encoding allows quantitative and qualitative comparisons of objects, supporting later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content, while also normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, such as calculating symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments like rows, grids, or repetition patterns. Consistent object IDs are enforced to prevent

relational ambiguity, and parsing splits output lines starting with parentheses, followed by automated consistency checks. This step transforms static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure allows reasoning over both pairwise interactions and more complex scene configurations, which is critical for tasks like symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. Grouping and Symmetry agent abstracts higher-order patterns from object lists and spatial relations, identifying clusters, rows, grids, and symmetries—including vertical, horizontal, rotational, and translational. Outputs are requested in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to maintain downstream compatibility. This agent captures emergent scene regularities that are not apparent at the object or pairwise relation level, providing critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent supports both perceptual inference and anticipatory reasoning for dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes including translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line using regex patterns that capture directional arrows, producing a temporal representation of the scene. This encoding allows the system to track not just static spatial configurations but also dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent ensures that transformations are linked to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format containing `objects`, `relations`, `groups`, `symmetries`, and `transformations`. Prompts enforce consistent object IDs and structured formatting, while JSON parsing failures revert to empty dictionaries to ensure pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent creates a comprehensive scene representation that is both human-interpretable and machine-readable. The resulting unified structure serves as a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern recognition questions. Inputs include objects, attributes, relations, groups, symmetries, and transformations. The model considers geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For our technical implementation, we use Gemini-2.5-Flash with a low temperature (0.1) to ensure deterministic outputs and perform multiple iterations for refinement. API calls are handled via OpenRouter, supporting both text and image inputs, while parsing of model responses relies on regex and JSON-based extraction to generate structured outputs. All interactions are logged in JSON format for reproducibility, and the final output is a unified JSON per image containing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition allows each specialized model to focus on a constrained sub-task, enhancing overall accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol remains identical to the standard CoT and CoT-SR setting, using 5 balanced samples per sub-category (seed 42), making a total of 150 samples.

The Gemini-2.5-Flash agent results show a clear category-specific trade-off. Orientation is the only category with consistent and substantial gains, improving by +8.00% in MCQ and +36.00% in open-ended evaluation. This margin indicates that agentic decomposition is most useful when the task can be reduced to sequential alignment of directional cues.

In contrast, most other categories have neutral or negative gains. Depth & Occlusion drops by -24.00% in open-ended evaluation, Spatial Navigation drops by -12.00%, and Relative Positioning is

Table 25: Agent performance comparison for Gemini-2.5-Flash. Gains report the agentic pipeline minus the normal baseline for MCQ and open-ended evaluation, making category-specific benefits and regressions explicit.

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67%	57.33%	+2.66%	35.83%	31.33%	-4.50%

unchanged in MCQ while declining by -14.00% in open-ended evaluation. These margins show that agent loops do not add robustness when the underlying depth, occlusion, or relational representation is weak.

Some categories show small positive MCQ effects, such as Size & Scale (+4.00%), but those gains do not transfer to open-ended evaluation. Even 3D Geometry declines by -4.00% in MCQ and -12.00% in open-ended evaluation. The agent comparison therefore reinforces the broader empirical conclusion: multi-step scaffolding can improve selected alignment tasks, but robust spatial reasoning requires deeper integration of geospatial understanding and perceptual representation.

Overall, the mixed results emphasize that agentic workflows alone are insufficient for advancing performance in this domain. While orientation tasks benefit from structured reasoning, most categories expose the models’ lack of inherent awareness of perceptual concepts. This makes errors propagate through multi-step reasoning, amplifying performance drops. Consequently, our benchmark serves as a strong stress test, highlighting that future progress will require embedding genuine perceptual and geospatial priors rather than relying solely on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section complements the quantitative results with qualitative evidence about where the measured errors come from. We define each recurring error class, connect it to diagnostic root causes, and describe verification protocols so the failure patterns can be interpreted alongside the category-level accuracy margins.

I.1 OVERVIEW

Across the sampled cases, models often recognize object categories, colors, and simple binary relations, but their accuracy drops when several spatial cues must be integrated. The failures cluster into interpretable classes rather than random noise: spatial mislocalization, perspective and scale mistakes, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, matching the quantitative finding that weaknesses are systematic across model families. In the following, we unpack each error class with representative examples and root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** When queries require distinguishing one object from a set of visually similar candidates, models frequently misidentify the target. For example, if asked to point out “the leftmost red cube among three stacked cubes,” the model may incorrectly output the middle cube, influenced

by lighting highlights or central positioning. This problem becomes particularly acute in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share the same color and shape. In such cases, the model’s errors are not random: it tends to privilege objects that appear most salient due to size, brightness, or centrality, even when these cues contradict the explicit spatial instructions in the query. These consistent patterns highlight that the failures are systematic, not accidental.

★ **Analysis.** The recurring mistakes suggest that models lack a robust object-centric representation, or what cognitive science would call an “index” that persists across reasoning steps. Instead of maintaining a stable pointer to the correct candidate throughout the reasoning chain, models appear to drift, often reassigning attention to whichever object seems most visually striking at intermediate layers. This leads to reasoning errors such as starting with the leftmost cube but ending the chain with the brightest or largest cube. The inability to anchor spatial reference consistently makes tasks involving relative comparisons, like “the ball behind the second chair”, particularly fragile. In essence, the model operates more like a salience detector than a referential reasoner.

⊗ **Root cause.** At the architectural level, feature pooling operations, whether via attention or convolution, tend to mix contextual information from neighboring regions. While this is beneficial for global context, it erases the fine-grained, object-specific identity needed to distinguish between similar candidates. Once pooled, features of two adjacent red cubes may be nearly indistinguishable in the latent space. Without an explicit mechanism, such as slot-based representations or pointer tokens, the network collapses distinct entities into blended embeddings. This structural limitation prevents reliable referent tracking across reasoning steps and explains why models consistently confuse dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks involving metric reasoning about relative size, fit, or scale are inconsistently handled. On simple comparisons like “is a car larger than a bicycle,” models typically succeed because such relationships align with strong statistical priors. However, when faced with near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” errors become frequent and unpredictable. A common pattern is reliance on canonical object sizes: for example, predicting that a door must be taller than a truck even in a rendering where the truck is scaled down. Another failure occurs in unusual visualizations, such as toy miniatures or architectural models, where familiar proportions are deliberately inverted. In these cases, models cling to their priors and ignore the actual geometric evidence visible in the image.

★ **Analysis.** These behaviors indicate that models rely heavily on distributional statistics of object categories rather than performing true geometric inference. Instead of reasoning about projection geometry, relative bounding box sizes, or depth cues, they substitute their world knowledge of typical scales. This shortcut works well in everyday contexts but fails catastrophically in edge cases that deviate from the training distribution. For instance, when an unusually small airplane is rendered next to a person, the model still predicts the airplane as larger. Such errors highlight the lack of grounding in pixel-level metric cues and the absence of consistent use of perspective or vanishing points.

⊗ **Root cause.** Current architectures do not include explicit geometric reasoning modules that preserve projection-aware relationships. Supervision for scale and fit tasks is limited, so models learn correlations between object category labels and size but not the underlying geometry. Furthermore, training objectives prioritize overall accuracy rather than fine-grained metric consistency, meaning that scale-sensitive errors are under-penalized. Without inductive biases for 3D reasoning or targeted supervision, the model defaults to memorized priors about canonical sizes rather than interpreting the actual perspective geometry in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object is in front, behind, or partially hidden remains a core weakness. Models often misreport the foreground object when two animals overlap, or when semi-transparent barriers like glass or fences are involved. For instance, in an image where a cat sits partially behind a curtain, models may incorrectly describe the curtain as being occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, are particularly error-prone: they are frequently omitted altogether, leading to wrong answers in questions like “what object lies behind the fence.” These mistakes occur consistently, especially when occlusion cues are subtle or when the depth difference is small.

★ **Analysis.** Such failures indicate that models treat depth and occlusion as weak, secondary signals. While the pixel data contains clear occlusion cues (e.g., edge boundaries, T-junctions), the intermediate features learned by the model do not retain them with sufficient resolution. Aggregation layers blur boundaries and downweight thin structures, effectively removing the very cues necessary to resolve ordering. This is why models succeed in gross depth distinctions (foreground vs. background) but collapse in fine-grained cases where partial occlusion must be inferred. In practice, they guess rather than infer, with systematic biases toward treating the visually larger or brighter object as foreground.

⊗ **Root cause.** The architecture lacks explicit mechanisms for multi-scale feature retention and has no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are easily lost when features are pooled across large receptive fields. Furthermore, training data often underrepresents subtle occlusion cases, especially those involving transparency or fine boundaries. As a result, the learned representations flatten layered scenes into 2D appearance maps, preventing robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models sometimes conflate perceptual attributes (such as color, texture, or material) with functional or semantic roles. A classic example is misjudging whether a crosswalk is safe: models may base the answer solely on the presence of painted stripes while ignoring the state of the traffic light or approaching vehicles. Similarly, a surface with a metallic texture may be misclassified as “steel” even if it is painted plastic. In tool-use contexts, the presence of handles or familiar shapes may lead the model to infer functionality that the object does not possess. These confusions are frequent in tasks requiring fine-grained visual grounding of attributes.

★ **Analysis.** These errors reveal that language priors strongly bias the model’s predictions when visual input is ambiguous. The decoder favors plausible narratives based on co-occurrence in the training data: “crosswalks are safe when stripes are visible,” “shiny textures correspond to metal,” and so on. This shortcut allows models to produce semantically coherent answers even when they do not inspect the relevant visual evidence. The result is superficially correct reasoning in familiar contexts but systematic errors in atypical or adversarial scenarios. In practice, this manifests as overfitting to textual stereotypes rather than grounding in perceptual details.

⊗ **Root cause.** The underlying training objective, maximizing likelihood of fluent and plausible text, does not enforce visual grounding. As long as the generated answer is semantically reasonable, the loss is minimized, even if the reasoning is incorrect. Consequently, the model is incentivized to substitute common-sense priors for genuine perceptual inference. Without auxiliary losses or verification steps to tie textual claims to image regions, models will continue to swap perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** For explanatory or why-style questions, models often generate coherent but unsupported narratives. For example, when asked “why is the object leaning,” a model might answer “because the surface is slippery” even though slipperiness cannot be visually inferred. In another case, a tilted shelf might be explained as “due to an earthquake,” a claim entirely outside the evidence in the image. These errors highlight a tendency to fill gaps with imaginative but ungrounded rationales, producing fluent explanations that mislead users about the true evidence.

★ **Analysis.** The issue stems from a decoupling between text generation and visual grounding. The model generates high-quality language that reads convincingly, but there is no mechanism ensuring that these rationales are tied to actual visual features. Confidence calibration compounds the problem: models often assign high certainty to fabricated explanations, giving the false impression of reliability. As a result, open-ended outputs may look authoritative while being factually baseless. This mode of failure is particularly concerning in safety-critical domains where plausible but false rationales could misguide decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence but not factual grounding. Decoders are optimized to produce the most probable continuation of text given the prompt, without constraints linking the explanation back to specific image evidence. The lack of an explicit verification layer or grounding requirement means that models default to linguistic plausibility over perceptual truth. Without additional architectural or training constraints,

open-ended rationalizations will continue to prioritize surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

When comparing failures across different models, several consistent patterns appear, revealing insights into the underlying mechanisms of errors. One clear trend is that disagreements between models are systematic rather than random: some models consistently rely on geometric cues such as relative size, depth ordering, and occlusion relationships, while others fall back on linguistic priors or common-sense assumptions derived from training text. This leads to predictable divergences, for example when two models give different answers about which object is “in front” despite seeing the same scene. Errors also spike when multiple cues must be integrated simultaneously. Tasks that require reasoning about occlusion, relative scale, and perspective together are particularly challenging; for instance, understanding whether a larger object partially occluding a smaller one is in front or behind often triggers consistent mispredictions. Calibration is another noticeable issue: high-confidence predictions frequently correspond to incorrect answers, especially in open-ended explanations or complex spatial queries. These patterns indicate that errors arise from how models balance perceptual grounding and linguistic plausibility. Different architectures exhibit distinct tendencies, some over-rely on visual geometry, others overfit to language correlations, so failures are not just isolated mistakes but reflect deeper trade-offs in reasoning strategies. This also implies that any attempt to improve performance should account for both sources of bias simultaneously rather than treating errors as independent events.

I.4 MITIGATION STRATEGIES

Given these error patterns, several targeted strategies can be pursued to reduce failures without requiring complete architectural redesign. For instance, geometry-aware supervision can be incorporated by adding auxiliary tasks like depth prediction, occlusion ordering, or overlap estimation. This encourages models to respect spatial constraints and improves consistency across predictions. Multi-scale feature retention is another approach: by using edge-aware pooling or attention mechanisms, models can maintain information from thin or partially occluded structures that would otherwise vanish in standard downsampling. Coordinate anchoring can help resolve confusion between viewer-centric and object-centric frames by explicitly representing reference points for relative positioning. Counterfactual augmentation can also be applied, where distractor objects are systematically swapped or perturbed, forcing the model to rely on the true referent cues rather than heuristics. Lightweight verification modules can be added on top of existing outputs, cross-checking claims against visual or geometric constraints to filter implausible predictions. Each of these strategies can be implemented incrementally, providing clear paths to reduce systematic failures while keeping the core model largely unchanged.

In addition to these architectural and training augmentations, supervised fine-tuning (SFT) offers a practical pathway to instill more reliable spatial reasoning. By curating datasets that explicitly highlight failure cases, such as near-threshold scale comparisons, occlusion ambiguities, or reference resolution traps, SFT can directly expose models to the edge scenarios where they otherwise default to heuristics. Carefully designed instruction-style fine-tuning further encourages step-by-step reasoning, prompting the model to verbalize intermediate spatial relations rather than skipping to a guess. This not only strengthens grounding but also improves interpretability, as errors become traceable to specific reasoning steps. Moreover, SFT can be paired with counterfactual or adversarial augmentations, ensuring that the model does not simply memorize but instead learns invariances to distractors and context shifts. Importantly, these interventions require no wholesale changes to the architecture: they adapt existing models by reshaping their inductive biases through supervision. In this sense, SFT serves as a lightweight yet powerful complement to geometry-aware and verification-based strategies.

In this work, we additionally performed SFT using 40% of the training data and observed clear performance gains, particularly on multiple-choice questions. This suggests that targeted supervision helps models better internalize spatial cues and resolve distractor confusions when answer options are well-defined. However, improvements on open-ended responses were more limited. The model continues to produce fluent but weakly grounded explanations, highlighting that SFT alone cannot fully address rationalization without verification. These findings reinforce the need to combine SFT

2754 with geometry-aware supervision, counterfactual augmentations, and lightweight verifiers to achieve
2755 balanced progress across both structured and free-form tasks.

2756 Our analysis shows that failures are not arbitrary but structured and interpretable, emerging from
2757 recurring patterns rather than random noise. This indicates that VLM errors follow systematic
2758 trajectories tied to the inductive biases of their architectures and training regimes. For example,
2759 geometric reasoning often collapses when occlusion cues are weak, and in such cases language
2760 priors take over, producing answers that sound plausible but lack visual grounding. Similarly, scale
2761 inference drifts when canonical object size priors dominate over pixel-level metric cues, leading to
2762 consistent misjudgments in near-threshold scenarios. These findings highlight that the gap is not
2763 simply one of capacity but of representation: current models lack explicit mechanisms to anchor
2764 visual features to grounded geometric relations.

2765 By adopting the proposed verification protocols and targeted mitigation strategies, researchers can
2766 address these weaknesses directly. Geometry-aware supervision and counterfactual augmentation
2767 would help models resist shortcuts, while multi-scale feature retention ensures thin structures and
2768 partial occlusions are not discarded. SFT, when carefully applied, can further strengthen structured
2769 reasoning by aligning model predictions with task constraints, though it must be combined with
2770 grounding objectives to avoid overfitting to discrete-choice formats. Importantly, verification modules
2771 provide a lightweight way to enforce consistency between claimed relations and image evidence,
2772 counteracting the tendency to rationalize without checking. Taken together, these steps chart a
2773 practical pathway toward systems that reason more like humans: robust to clutter, interpretable in
2774 their failures, and capable of explaining not only what they predict but why.

2775
2776
2777
2778
2779
2780
2781
2782
2783
2784
2785
2786
2787
2788
2789
2790
2791
2792
2793
2794
2795
2796
2797
2798
2799
2800
2801
2802
2803
2804
2805
2806
2807

2808
2809
2810
2811
2812
2813
2814
2815
2816
2817
2818
2819
2820
2821
2822
2823
2824
2825
2826
2827
2828
2829
2830
2831
2832
2833
2834
2835
2836
2837
2838
2839
2840
2841
2842
2843
2844
2845
2846
2847
2848
2849
2850
2851
2852
2853
2854
2855
2856
2857
2858
2859
2860
2861

Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
JSON array, e.g.:
[
 {{"ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position":
"Leftmost"}},
 {{"ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position":
"Center"}}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

2862
2863
2864
2865
2866
2867
2868
2869
2870
2871
2872
2873
2874
2875
2876
2877
2878
2879
2880
2881
2882
2883
2884
2885
2886
2887
2888
2889
2890
2891
2892
2893
2894
2895
2896
2897
2898
2899
2900
2901
2902
2903
2904
2905
2906
2907
2908
2909
2910
2911
2912
2913
2914
2915

Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

2916
2917
2918
2919
2920
2921
2922
2923
2924
2925
2926
2927
2928
2929
2930
2931
2932
2933
2934
2935
2936
2937
2938
2939
2940
2941
2942
2943
2944
2945
2946
2947
2948
2949
2950
2951
2952
2953
2954
2955
2956
2957
2958
2959
2960
2961
2962
2963
2964
2965
2966
2967
2968
2969

Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]
}
```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark samples for all 30 sub-categories in SPATIALAB, illustrating the diverse spatial reasoning challenges covered. These samples highlight the variety of visual phenomena, including 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation tasks. Together, they provide a concrete view of the types of reasoning each model must perform to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Representative benchmark examples (3D Geometry: Gravity Effects) with evaluation results.

3024
3025
3026
3027
3028
3029
3030
3031
3032
3033
3034
3035
3036
3037
3038
3039
3040
3041
3042
3043
3044
3045
3046
3047
3048
3049
3050
3051
3052
3053
3054
3055
3056
3057
3058
3059
3060
3061
3062
3063
3064
3065
3066
3067
3068
3069
3070
3071
3072
3073
3074
3075
3076
3077

3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Representative benchmark examples (3D Geometry: Shape Projection) with evaluation results.

3078
3079
3080
3081
3082
3083
3084
3085
3086
3087
3088
3089
3090
3091
3092
3093
3094
3095
3096
3097
3098
3099
3100
3101
3102
3103
3104
3105
3106
3107
3108
3109
3110
3111
3112
3113
3114
3115
3116
3117
3118
3119
3120
3121
3122
3123
3124
3125
3126
3127
3128
3129
3130
3131

3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Representative benchmark examples (3D Geometry: Spatial Containment) with evaluation results.

3132
3133
3134
3135
3136
3137
3138
3139
3140
3141
3142
3143
3144
3145
3146
3147
3148
3149
3150
3151
3152
3153
3154
3155
3156
3157
3158
3159
3160
3161
3162
3163
3164
3165
3166
3167
3168
3169
3170
3171
3172
3173
3174
3175
3176
3177
3178
3179
3180
3181
3182
3183
3184
3185

3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Representative benchmark examples (3D Geometry: Volume Comparison) with evaluation results.

3186
3187
3188
3189
3190
3191
3192
3193
3194
3195
3196
3197
3198
3199
3200
3201
3202
3203
3204
3205
3206
3207
3208
3209
3210
3211
3212
3213
3214
3215
3216
3217
3218
3219
3220
3221
3222
3223
3224
3225
3226
3227
3228
3229
3230
3231
3232
3233
3234
3235
3236
3237
3238
3239

3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- Gemini-2.5-Flash: The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- InternVL3.5-72B: From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Representative benchmark examples (3D Geometry: Shape Projection) with evaluation results.

3240
3241
3242
3243
3244
3245
3246
3247
3248
3249
3250
3251
3252
3253
3254
3255
3256
3257
3258
3259
3260
3261
3262
3263
3264
3265
3266
3267
3268
3269
3270
3271
3272
3273
3274
3275
3276
3277
3278
3279
3280
3281
3282
3283
3284
3285
3286
3287
3288
3289
3290
3291
3292
3293

Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Representative benchmark examples (Spatial Navigation: Accessibility Constraints) with evaluation results.

3294
3295
3296
3297
3298
3299
3300
3301
3302
3303
3304
3305
3306
3307
3308
3309
3310
3311
3312
3313
3314
3315
3316
3317
3318
3319
3320
3321
3322
3323
3324
3325
3326
3327
3328
3329
3330
3331
3332
3333
3334
3335
3336
3337
3338
3339
3340
3341
3342
3343
3344
3345
3346
3347

Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

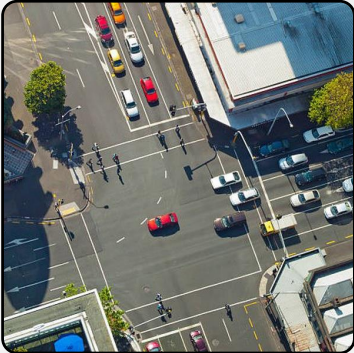
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Representative benchmark examples (Spatial Navigation: Obstacle Avoidance) with evaluation results.

3348
3349
3350
3351
3352
3353
3354
3355
3356
3357
3358
3359
3360
3361
3362
3363
3364
3365
3366
3367
3368
3369
3370
3371
3372
3373
3374
3375
3376
3377
3378
3379
3380
3381
3382
3383
3384
3385
3386
3387
3388
3389
3390
3391
3392
3393
3394
3395
3396
3397
3398
3399
3400
3401

Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- **Gemini-2.5-Flash:** The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- **InternVL3.5-72B:** The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Representative benchmark examples (Spatial Navigation: Pathway Existence) with evaluation results.

3402
3403
3404
3405
3406
3407
3408
3409
3410
3411
3412
3413
3414
3415
3416
3417
3418
3419
3420
3421
3422
3423
3424
3425
3426
3427
3428
3429
3430
3431
3432
3433
3434
3435
3436
3437
3438
3439
3440
3441
3442
3443
3444
3445
3446
3447
3448
3449
3450
3451
3452
3453
3454
3455

Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Representative benchmark examples (Spatial Navigation: Viewpoint Visibility) with evaluation results.

3456
3457
3458
3459
3460
3461
3462
3463
3464
3465
3466
3467
3468
3469
3470
3471
3472
3473
3474
3475
3476
3477
3478
3479
3480
3481
3482
3483
3484
3485
3486
3487
3488
3489
3490
3491
3492
3493
3494
3495
3496
3497
3498
3499
3500
3501
3502
3503
3504
3505
3506
3507
3508
3509

Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Representative benchmark examples (Spatial Navigation: Spatial Sequence) with evaluation results.

3510
3511
3512
3513
3514
3515
3516
3517
3518
3519
3520
3521
3522
3523
3524
3525
3526
3527
3528
3529
3530
3531
3532
3533
3534
3535
3536
3537
3538
3539
3540
3541
3542
3543
3544
3545
3546
3547
3548
3549
3550
3551
3552
3553
3554
3555
3556
3557
3558
3559
3560
3561
3562
3563

Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Representative benchmark examples (Size and Scale: Scale Consistency) with evaluation results.

3564
3565
3566
3567
3568
3569
3570
3571
3572
3573
3574
3575
3576
3577
3578
3579
3580
3581
3582
3583
3584
3585
3586
3587
3588
3589
3590
3591
3592
3593
3594
3595
3596
3597
3598
3599
3600
3601
3602
3603
3604
3605
3606
3607
3608
3609
3610
3611
3612
3613
3614
3615
3616
3617

Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Representative benchmark examples (Size and Scale: Shadow-Size Projection) with evaluation results.

3618
3619
3620
3621
3622
3623
3624
3625
3626
3627
3628
3629
3630
3631
3632
3633
3634
3635
3636
3637
3638
3639
3640
3641
3642
3643
3644
3645
3646
3647
3648
3649
3650
3651
3652
3653
3654
3655
3656
3657
3658
3659
3660
3661
3662
3663
3664
3665
3666
3667
3668
3669
3670
3671

Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Representative benchmark examples (Size and Scale: Perspective Distortion) with evaluation results.

3672
3673
3674
3675
3676
3677
3678
3679
3680
3681
3682
3683
3684
3685
3686
3687
3688
3689
3690
3691
3692
3693
3694
3695
3696
3697
3698
3699
3700
3701
3702
3703
3704
3705
3706
3707
3708
3709
3710
3711
3712
3713
3714
3715
3716
3717
3718
3719
3720
3721
3722
3723
3724
3725

Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- **Gemini-2.5-Flash:** The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- **InternVL3.5-72B:** The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Representative benchmark examples (Size and Scale: Relative Size Comparison) with evaluation results.

3726
3727
3728
3729
3730
3731
3732
3733
3734
3735
3736
3737
3738
3739
3740
3741
3742
3743
3744
3745
3746
3747
3748
3749
3750
3751
3752
3753
3754
3755
3756
3757
3758
3759
3760
3761
3762
3763
3764
3765
3766
3767
3768
3769
3770
3771
3772
3773
3774
3775
3776
3777
3778
3779

Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Representative benchmark examples (Size and Scale: Relative Size Comparison) with evaluation results.

3780
3781
3782
3783
3784
3785
3786
3787
3788
3789
3790
3791
3792
3793
3794
3795
3796
3797
3798
3799
3800
3801
3802
3803
3804
3805
3806
3807
3808
3809
3810
3811
3812
3813
3814
3815
3816
3817
3818
3819
3820
3821
3822
3823
3824
3825
3826
3827
3828
3829
3830
3831
3832
3833

Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- Gemini-2.5-Flash: The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- InternVL3.5-72B: The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- Kimi-VL-A3B-Thinking-2506: The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Representative benchmark examples (Depth and Occlusion: Complete Occlusion Inference) with evaluation results.

3834
3835
3836
3837
3838
3839
3840
3841
3842
3843
3844
3845
3846
3847
3848
3849
3850
3851
3852
3853
3854
3855
3856
3857
3858
3859
3860
3861
3862
3863
3864
3865
3866
3867
3868
3869
3870
3871
3872
3873
3874
3875
3876
3877
3878
3879
3880
3881
3882
3883
3884
3885
3886
3887

Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Representative benchmark examples (Depth and Occlusion: Layering Order) with evaluation results.

3888
3889
3890
3891
3892
3893
3894
3895
3896
3897
3898
3899
3900
3901
3902
3903
3904
3905
3906
3907
3908
3909
3910
3911
3912
3913
3914
3915
3916
3917
3918
3919
3920
3921
3922
3923
3924
3925
3926
3927
3928
3929
3930
3931
3932
3933
3934
3935
3936
3937
3938
3939
3940
3941

Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Representative benchmark examples (Depth and Occlusion: Partial Occlusion) with evaluation results.

3942
3943
3944
3945
3946
3947
3948
3949
3950
3951
3952
3953
3954
3955
3956
3957
3958
3959
3960
3961
3962
3963
3964
3965
3966
3967
3968
3969
3970
3971
3972
3973
3974
3975
3976
3977
3978
3979
3980
3981
3982
3983
3984
3985
3986
3987
3988
3989
3990
3991
3992
3993
3994
3995

Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Representative benchmark examples (Depth and Occlusion: Reflective Surfaces) with evaluation results.

3996
3997
3998
3999
4000
4001
4002
4003
4004
4005
4006
4007
4008
4009
4010
4011
4012
4013
4014
4015
4016
4017
4018
4019
4020
4021
4022
4023
4024
4025
4026
4027
4028
4029
4030
4031
4032
4033
4034
4035
4036
4037
4038
4039
4040
4041
4042
4043
4044
4045
4046
4047
4048
4049

Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Representative benchmark examples (Depth and Occlusion: Transparency Effects) with evaluation results.

4050
4051
4052
4053
4054
4055
4056
4057
4058
4059
4060
4061
4062
4063
4064
4065
4066
4067
4068
4069
4070
4071
4072
4073
4074
4075
4076
4077
4078
4079
4080
4081
4082
4083
4084
4085
4086
4087
4088
4089
4090
4091
4092
4093
4094
4095
4096
4097
4098
4099
4100
4101
4102
4103

Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Representative benchmark examples (Orientation: Cardinal Direction) with evaluation results.

4104
4105
4106
4107
4108
4109
4110
4111
4112
4113
4114
4115
4116
4117
4118
4119
4120
4121
4122
4123
4124
4125
4126
4127
4128
4129
4130
4131
4132
4133
4134
4135
4136
4137
4138
4139
4140
4141
4142
4143
4144
4145
4146
4147
4148
4149
4150
4151
4152
4153
4154
4155
4156
4157

Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

- Up
- Down
- Away
- Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash: The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B: If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506: The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Representative benchmark examples (Orientation: Facing Direction) with evaluation results.

4158
4159
4160
4161
4162
4163
4164
4165
4166
4167
4168
4169
4170
4171
4172
4173
4174
4175
4176
4177
4178
4179
4180
4181
4182
4183
4184
4185
4186
4187
4188
4189
4190
4191
4192
4193
4194
4195
4196
4197
4198
4199
4200
4201
4202
4203
4204
4205
4206
4207
4208
4209
4210
4211

Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Representative benchmark examples (Orientation: Object Rotation) with evaluation results.

4212
4213
4214
4215
4216
4217
4218
4219
4220
4221
4222
4223
4224
4225
4226
4227
4228
4229
4230
4231
4232
4233
4234
4235
4236
4237
4238
4239
4240
4241
4242
4243
4244
4245
4246
4247
4248
4249
4250
4251
4252
4253
4254
4255
4256
4257
4258
4259
4260
4261
4262
4263
4264
4265

Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Representative benchmark examples (Orientation: Stacking Orientation) with evaluation results.

4266
4267
4268
4269
4270
4271
4272
4273
4274
4275
4276
4277
4278
4279
4280
4281
4282
4283
4284
4285
4286
4287
4288
4289
4290
4291
4292
4293
4294
4295
4296
4297
4298
4299
4300
4301
4302
4303
4304
4305
4306
4307
4308
4309
4310
4311
4312
4313
4314
4315
4316
4317
4318
4319

Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- **GPT-5-mini:** 2
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Representative benchmark examples (Orientation: Tool Handedness) with evaluation results.

4320
4321
4322
4323
4324
4325
4326
4327
4328
4329
4330
4331
4332
4333
4334
4335
4336
4337
4338
4339
4340
4341
4342
4343
4344
4345
4346
4347
4348
4349
4350
4351
4352
4353
4354
4355
4356
4357
4358
4359
4360
4361
4362
4363
4364
4365
4366
4367
4368
4369
4370
4371
4372
4373

Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Representative benchmark examples (Relative Positioning: Alignment Patterns) with evaluation results.

4374
4375
4376
4377
4378
4379
4380
4381
4382
4383
4384
4385
4386
4387
4388
4389
4390
4391
4392
4393
4394
4395
4396
4397
4398
4399
4400
4401
4402
4403
4404
4405
4406
4407
4408
4409
4410
4411
4412
4413
4414
4415
4416
4417
4418
4419
4420
4421
4422
4423
4424
4425
4426
4427

Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Representative benchmark examples (Relative Positioning: Betweenness Relationships) with evaluation results.

4428
4429
4430
4431
4432
4433
4434
4435
4436
4437
4438
4439
4440
4441
4442
4443
4444
4445
4446
4447
4448
4449
4450
4451
4452
4453
4454
4455
4456
4457
4458
4459
4460
4461
4462
4463
4464
4465
4466
4467
4468
4469
4470
4471
4472
4473
4474
4475
4476
4477
4478
4479
4480
4481

Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Representative benchmark examples (Relative Positioning: Corner/Angle Positioning) with evaluation results.

4482
4483
4484
4485
4486
4487
4488
4489
4490
4491
4492
4493
4494
4495
4496
4497
4498
4499
4500
4501
4502
4503
4504
4505
4506
4507
4508
4509
4510
4511
4512
4513
4514
4515
4516
4517
4518
4519
4520
4521
4522
4523
4524
4525
4526
4527
4528
4529
4530
4531
4532
4533
4534
4535

Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Saffari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Representative benchmark examples (Relative Positioning: Directional Relations) with evaluation results.

84

4536
4537
4538
4539
4540
4541
4542
4543
4544
4545
4546
4547
4548
4549
4550
4551
4552
4553
4554
4555
4556
4557
4558
4559
4560
4561
4562
4563
4564
4565
4566
4567
4568
4569
4570
4571
4572
4573
4574
4575
4576
4577
4578
4579
4580
4581
4582
4583
4584
4585
4586
4587
4588
4589

Relative Positioning : Proximity Gradients



Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- Gemini-2.5-Flash: "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- InternVL3.5-72B: The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- Kimi-VL-A3B-Thinking-2506: Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Representative benchmark examples (Relative Positioning: Proximity Gradients) with evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

To ensure that the quantitative comparisons are not artifacts of random initialization or dataset noise, we evaluate both model robustness across runs and dataset stability across repeated trials. These analyses are especially important for fine-grained spatial benchmarks, where small fluctuations could otherwise distort category-level margins (Drummond, 2009; Drummond & Japkowicz, 2010).

Let N denote the number of items *questions* in the benchmark and $R = 3$ the number of independent runs of the same model. Each run uses a distinct random seed while preserving all other hyperparameters.

K.1 MODEL ROBUSTNESS

We assess model-evaluation robustness with several complementary statistics, each designed to test whether reported accuracy differences remain stable across repeated runs.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify agreement between runs, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979), using a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ is the variance between items and σ_{within}^2 is the residual variance across runs. ICC values above 0.75 indicate good reliability, while values above 0.9 imply excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We also analyze the stability of the dataset itself, whether question difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

To further assess **robustness**, we perform a resampling analysis. From the total of $C = 30$ subcategories, we randomly sample subsets of size $S \in 20, 25$ for each sub-category from each of the $R = 3$ runs, across both models and evaluation types (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary check, the Wilcoxon signed-rank test assesses median differences between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ be the difference in accuracies for run r , excluding ties ($d_r \neq 0$). The test statistic is computed as:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

If the resulting p -value satisfies $p > 0.05$, we again fail to reject H_0 , confirming that the accuracies from 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty can be defined as $d_i = 1 - \bar{a} * i$. If item difficulty is consistent, the correlation between runs should be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) suggests stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

To measure overall internal consistency, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 is the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ indicate excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We perform the analysis on two models: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**, thereby covering both large- and small-scale models as well as proprietary and open-source paradigms. We also conduct all tests in both multiple-choice (MCQ) and open-ended evaluation formats (treated as binary: correct or incorrect). For statistical analysis, we used the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for `wilcoxon` test; `pingouin`³ (Vallat, 2018) for `intraclass_correlation` and `cronbach_alpha`; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We analyzed the stability of model accuracy across $R = 3$ independent runs. The standard deviation of accuracy (σ_A) across runs is negligible ($< 0.6\%$), indicating that the reported scores are precise estimates unaffected by random seeding noise.

Table 26: Mean accuracy and run-to-run variance across 3 independent runs. Low standard deviations indicate stable estimates for both MCQ and open-ended formats.

Model	Format	Mean Accuracy ($\bar{A}$)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The consistently low standard deviations across all models and evaluation formats indicate that performance estimates are highly stable and reproducible, with minimal sensitivity to stochastic run-to-run variation. This stability suggests that observed accuracy differences meaningfully reflect underlying model capability rather than noise introduced by random initialization or decoding. Notably, although open-ended evaluation exhibits slightly higher variance than MCQ, the absolute magnitude remains negligible, confirming that free-form generation is also robustly assessed under the current protocol. These findings imply that the dataset size and scoring methodology are sufficient to yield convergent accuracy estimates without requiring extensive repeated runs. Consequently, single-run evaluations are likely to be reliable for large-scale comparisons, while multi-run analyses primarily serve to validate robustness. Overall, this strengthens the empirical credibility of the reported results and supports confident cross-model and cross-format performance interpretation.

Intra-Class Correlation (ICC)

We computed ICC(3,k) to quantify the reliability of the scoring mechanism.

Table 27: Intra-Class Correlation (ICC) reliability scores for repeated model evaluations. Values near 1 indicate stable item-level ordering across runs.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The consistently high ICC(3,k) scores demonstrate that the evaluation exhibits near-perfect reliability across independent runs, regardless of model size or response format. This indicates that item-level difficulty ordering is preserved, and that performance fluctuations do not alter relative rankings among samples. Such strong agreement suggests that the scoring pipeline is internally consistent and

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

not sensitive to stochastic variation in model outputs. Importantly, this rules out random guessing as a dominant factor, even in open-ended generation, where variability is typically higher. From an evaluation standpoint, these results imply that the benchmark provides a dependable signal for assessing model capability. Consequently, longitudinal comparisons and fine-grained ablations can be conducted with high confidence in measurement stability.

Overall, the combined evidence from multiple-run variance analysis and ICC reliability demonstrates that the evaluation is highly robust to stochastic effects. Both absolute performance (low standard deviation) and relative performance (near-perfect ICC) remain stable across independent runs, confirming that results are not driven by random seeding or decoding noise. This robustness ensures that observed performance gaps across models and formats reflect genuine capability differences. As a result, the evaluation framework provides a reliable and reproducible basis for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To ensure statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, each time drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling stability over 1,000 trials using a Wilcoxon signed-rank test ($S = 20$ vs. $S = 25$). The table reports whether smaller subcategory samples preserve comparable accuracy estimates.

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-vl-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-vl-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results provide strong empirical evidence that accuracy estimates have already converged at relatively small subcategory sizes. The consistently high mean (p)-values and the absence of statistically significant differences across all 1,000 trials indicate that performance metrics are insensitive to moderate reductions in sample size. This stability holds uniformly across models and evaluation formats, suggesting that neither model scale nor output structure introduces additional sampling uncertainty. Importantly, these findings imply that the variance in accuracy is dominated by model behavior rather than data sparsity at the subcategory level. As a result, the current subcategory size (≈ 50) can be considered statistically redundant, offering a substantial safety margin beyond the minimum required for reliable estimation. This supports the validity of the benchmark design and suggests that future extensions could reallocate samples across categories without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measured the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s Alpha to assess internal consistency.

Table 29: Item-level consistency metrics across repeated runs. Mean pairwise correlation and Cronbach’s α quantify whether item difficulty and aggregate scores remain stable.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-vl-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-vl-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The extremely high mean pairwise correlations (> 0.94) demonstrate strong item-level stability, indicating that the relative difficulty of individual questions is preserved across independent runs. This consistency shows that model responses are systematically structured rather than subject to

random variation, even under stochastic decoding. The similarly high Cronbach’s α values (> 0.9) further confirm that the dataset exhibits excellent internal coherence, with items contributing reliably to the overall accuracy measure. Notably, the slight reductions observed in open-ended formats are expected given their greater expressive flexibility, yet remain well within the range of excellent reliability. Together, these results indicate that the benchmark functions as a unified and internally consistent measurement instrument. Consequently, model performance differences can be confidently attributed to true capability variation rather than instability at the item level.

K.3.3 OVERALL DISCUSSION

The robustness analyses support the reliability of the reported comparisons. ICC and Cronbach’s α scores consistently exceed 0.98, indicating near-perfect reliability and internal consistency, while the standard deviation of accuracy across runs remains negligible ($< 0.6\%$). The resampling study further shows that smaller subsets ($N = 20$ and $N = 25$) produce statistically indistinguishable estimates, confirming that the chosen subcategory size (~ 25) is sufficient for stable measurement. Together, these results indicate that both absolute accuracy and relative model rankings are stable across runs and sampling regimes, so the fine-grained margins reported in the paper reflect model capability differences rather than evaluation noise.